# TEMPEST: Porewater ICP

2023-2025 Samples

2026-04-28

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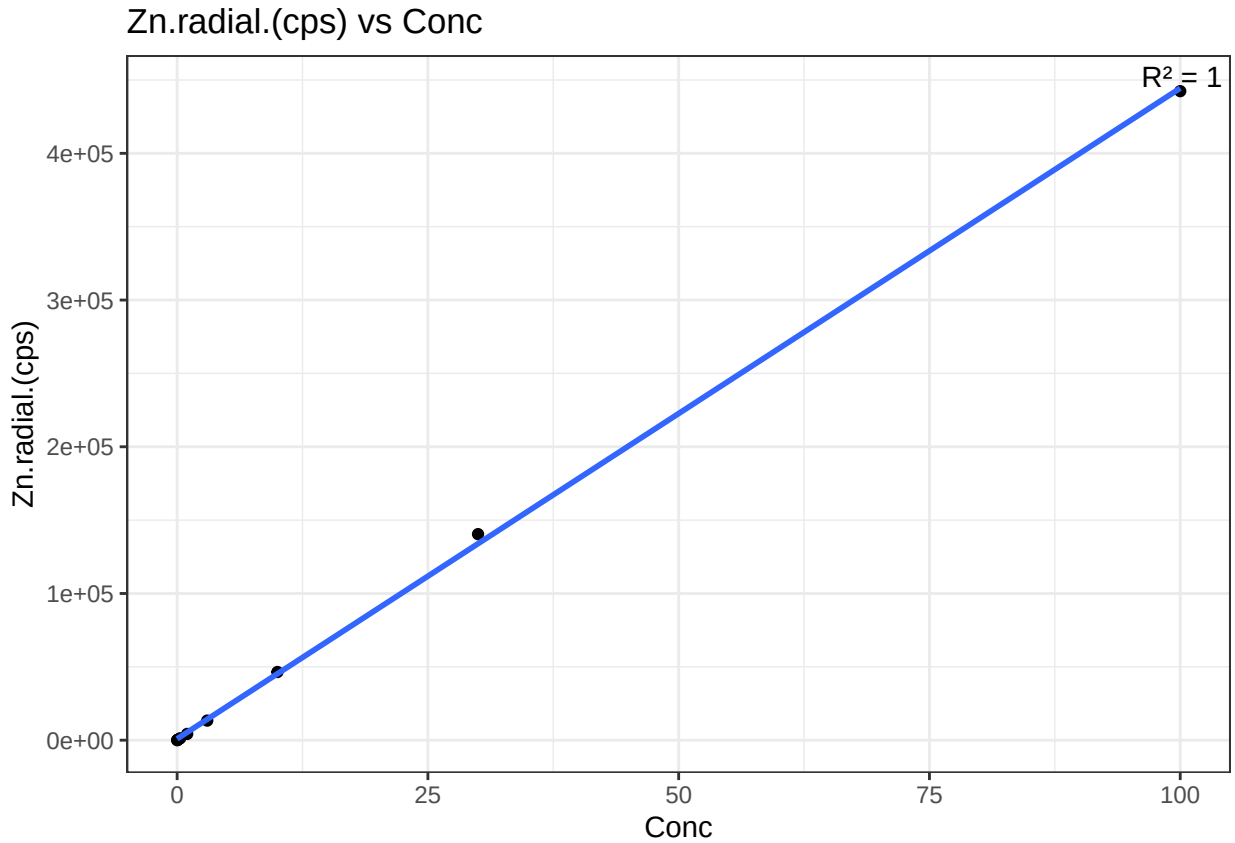
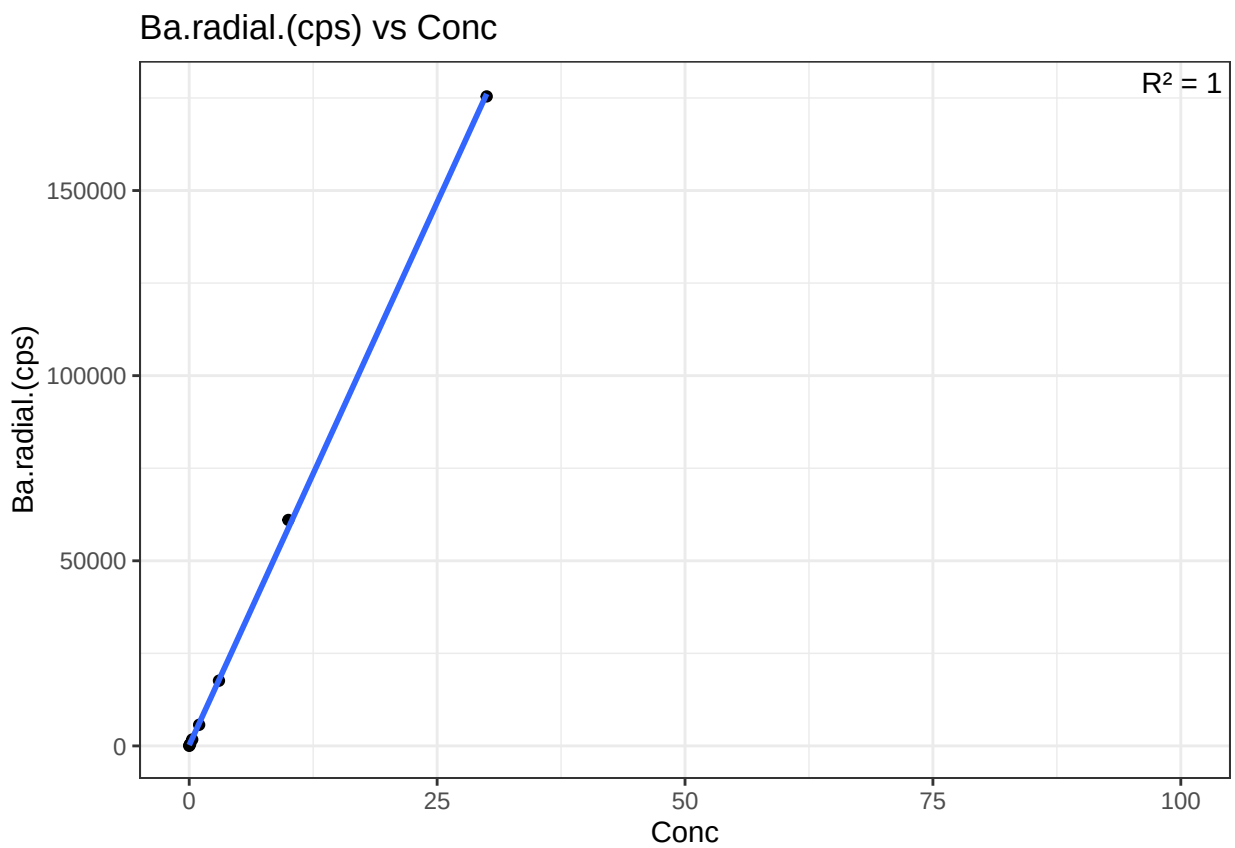
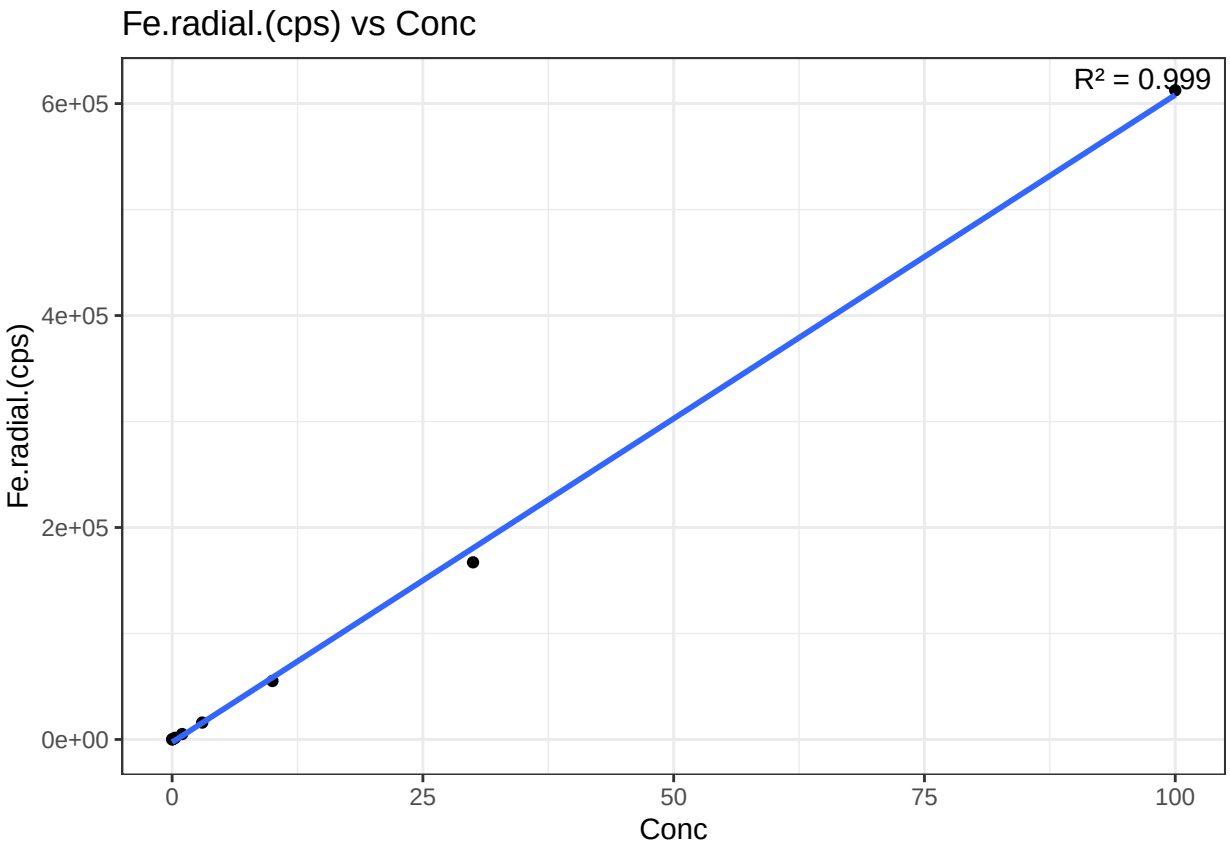
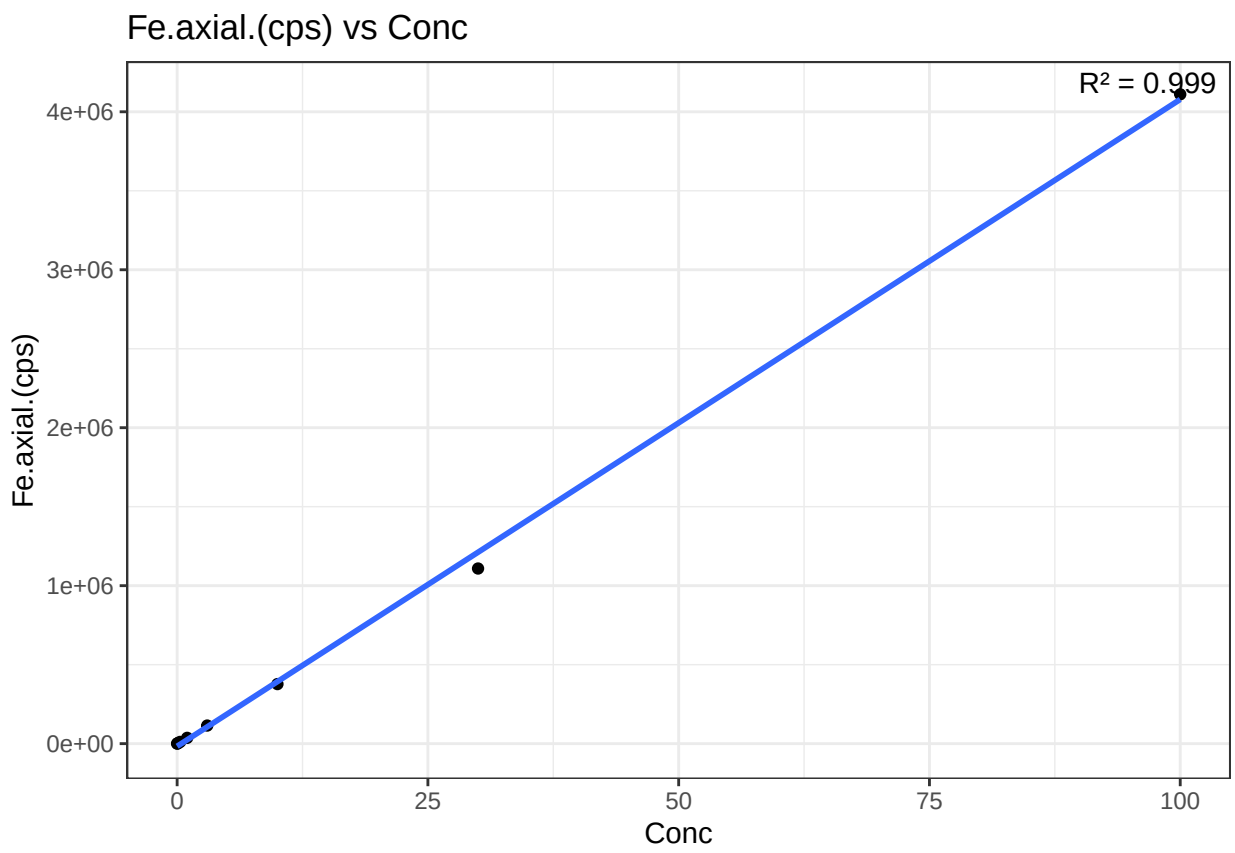
##Add Required Packages

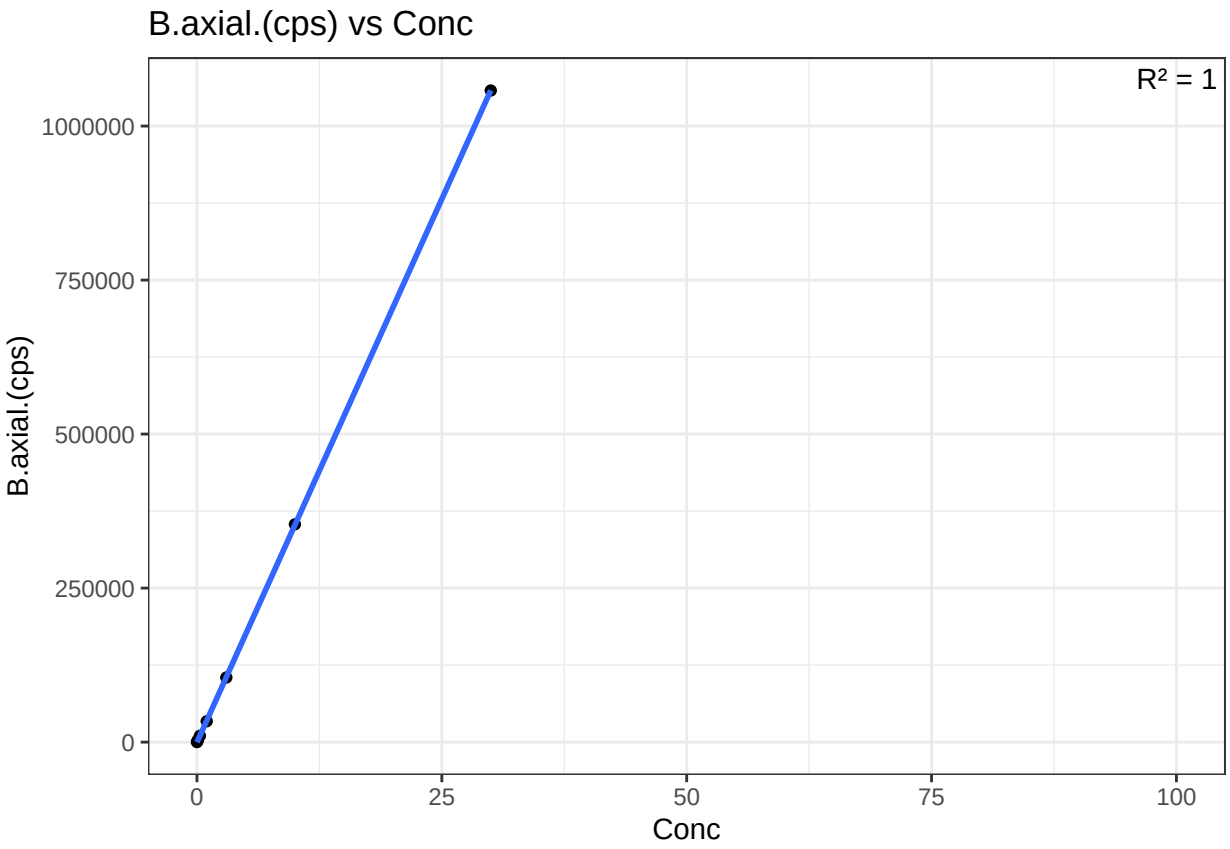
## 0.1 Import and QAQC Metal Standards

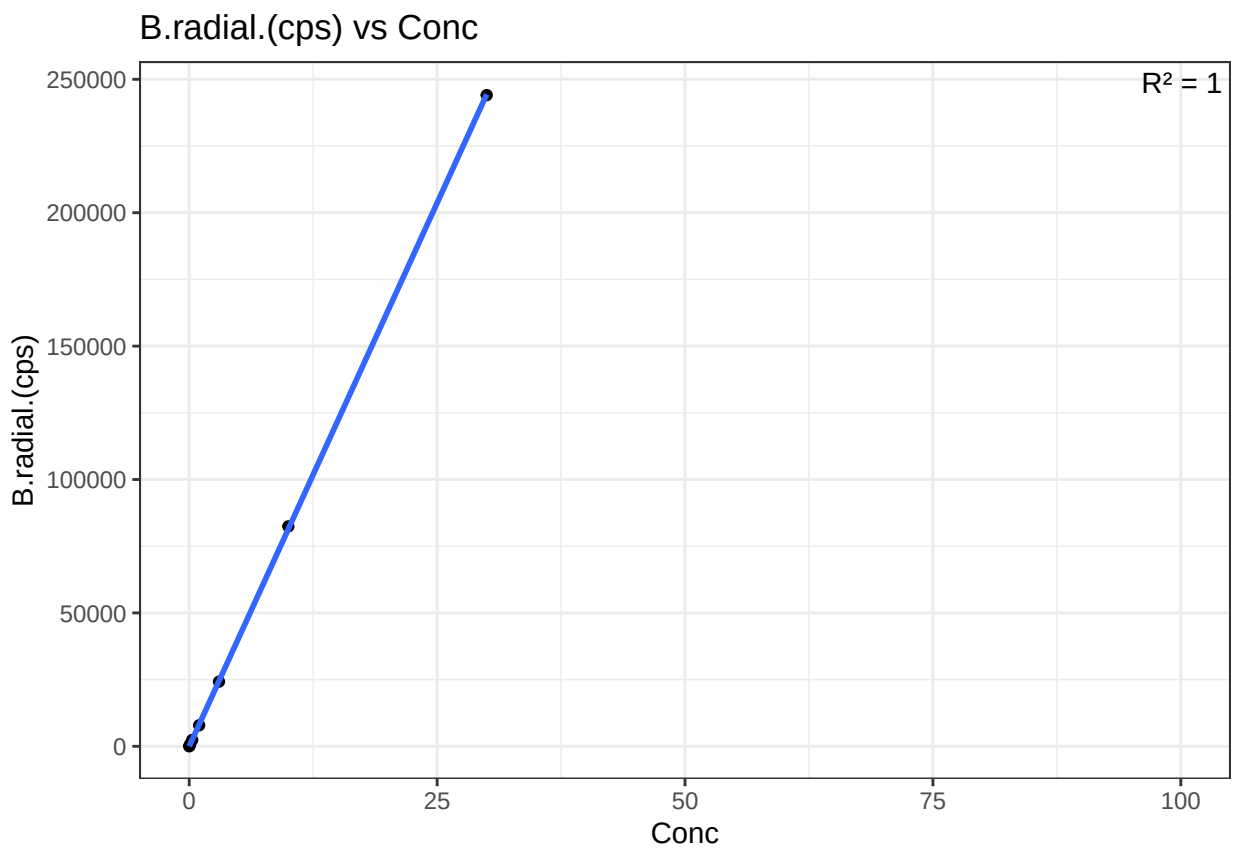


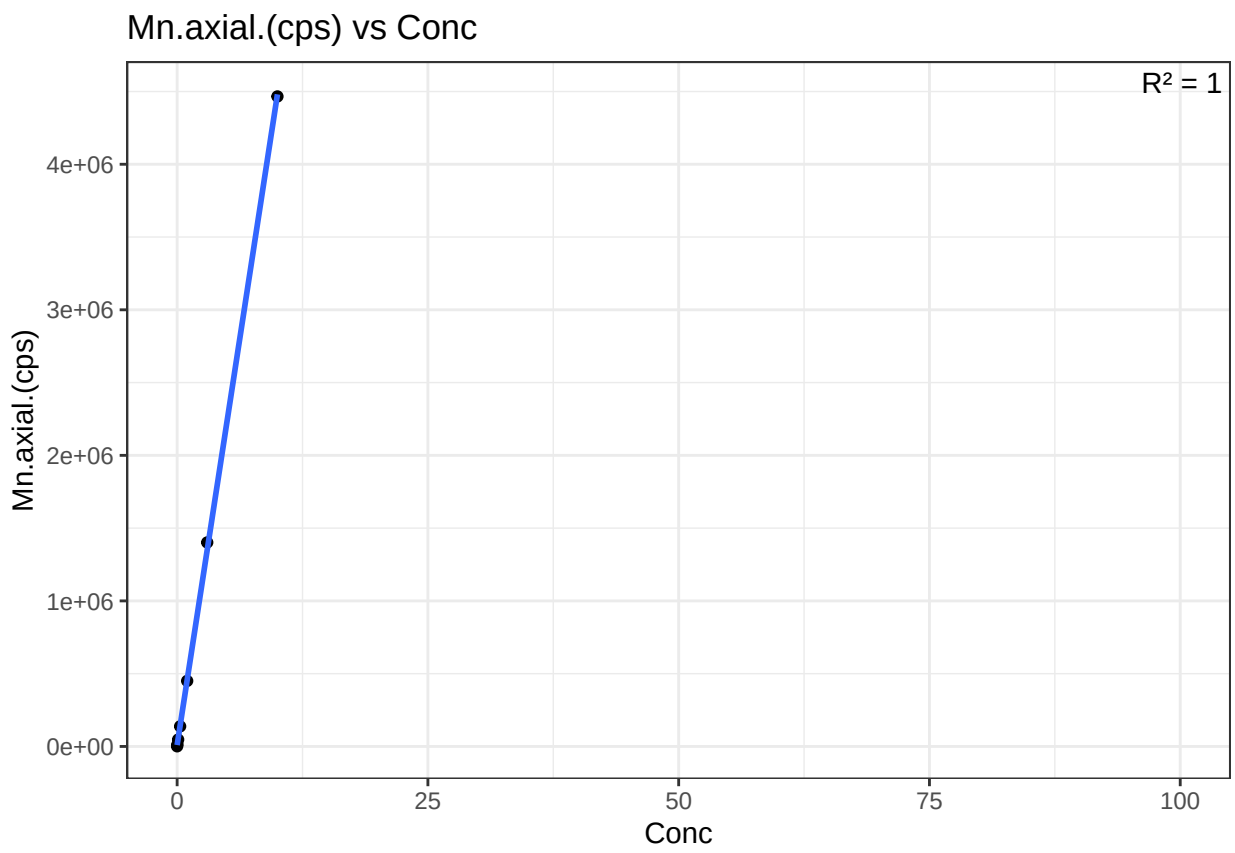



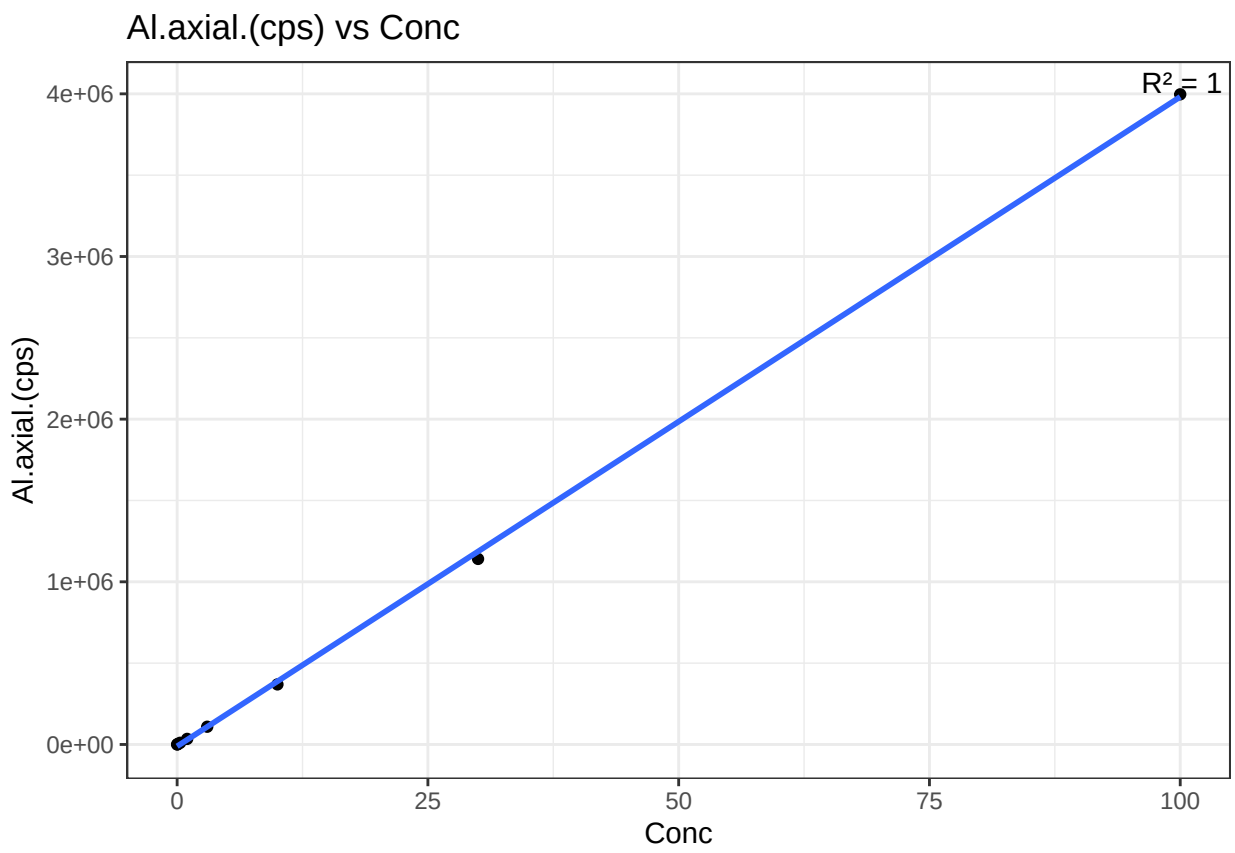

Fe.radial.(cps) vs Conc
R² = 0.999
6e+05
4e+05
2e+05
0e+00
Fe.radial.(cps)
0
25
50
75
100
Conc

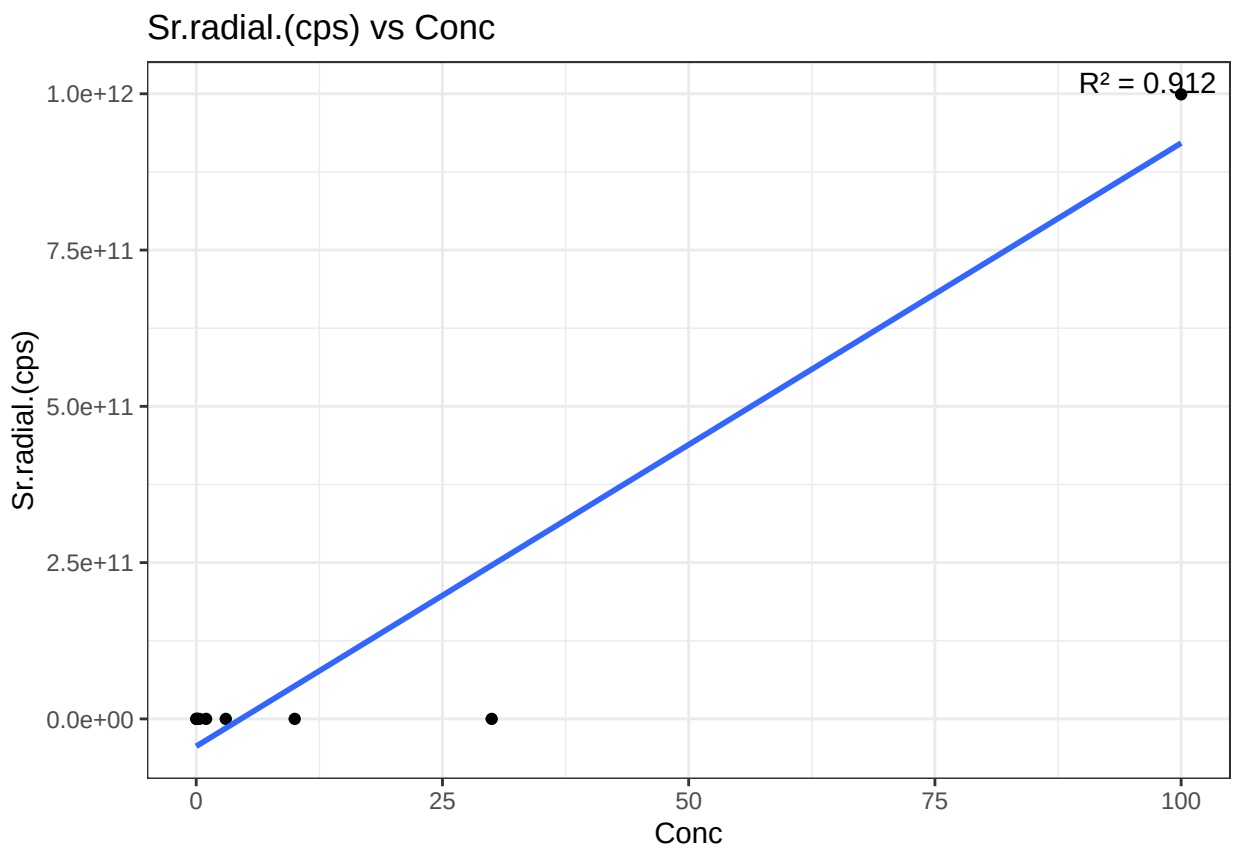

Fe.axial.(cps) vs Conc
R² = 0.999
4e+06
3e+06
2e+06
1e+06
0e+00
Fe.axial.(cps)
0
25
50
75
100
Conc

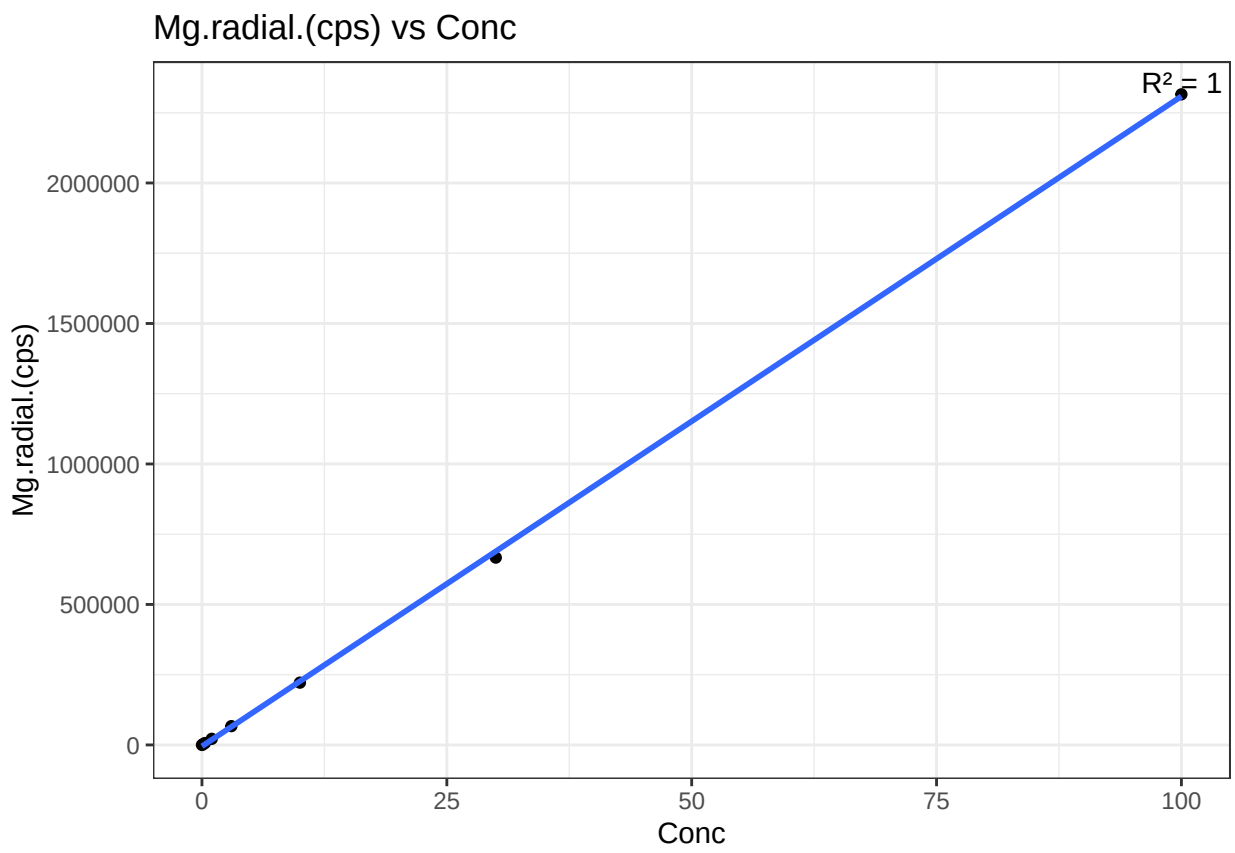

B.axial.(cps) vs Conc
R² = 1
1000000
750000
500000
250000
0
B.axial.(cps)
0
25
50
75
100
Conc

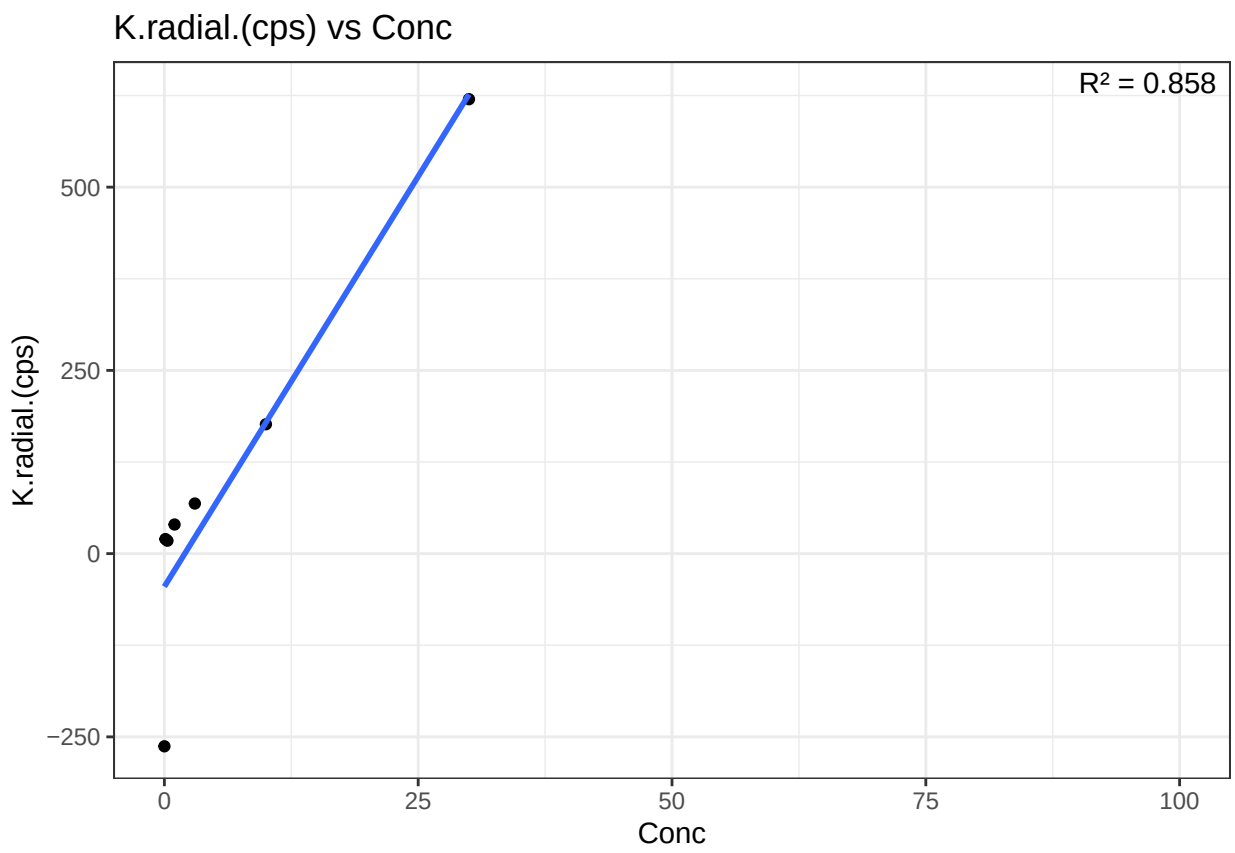

B.radial.(cps) vs Conc
R² = 1
250000
200000
150000
100000
50000
0
B.radial.(cps)
0
25
50
75
100
Conc

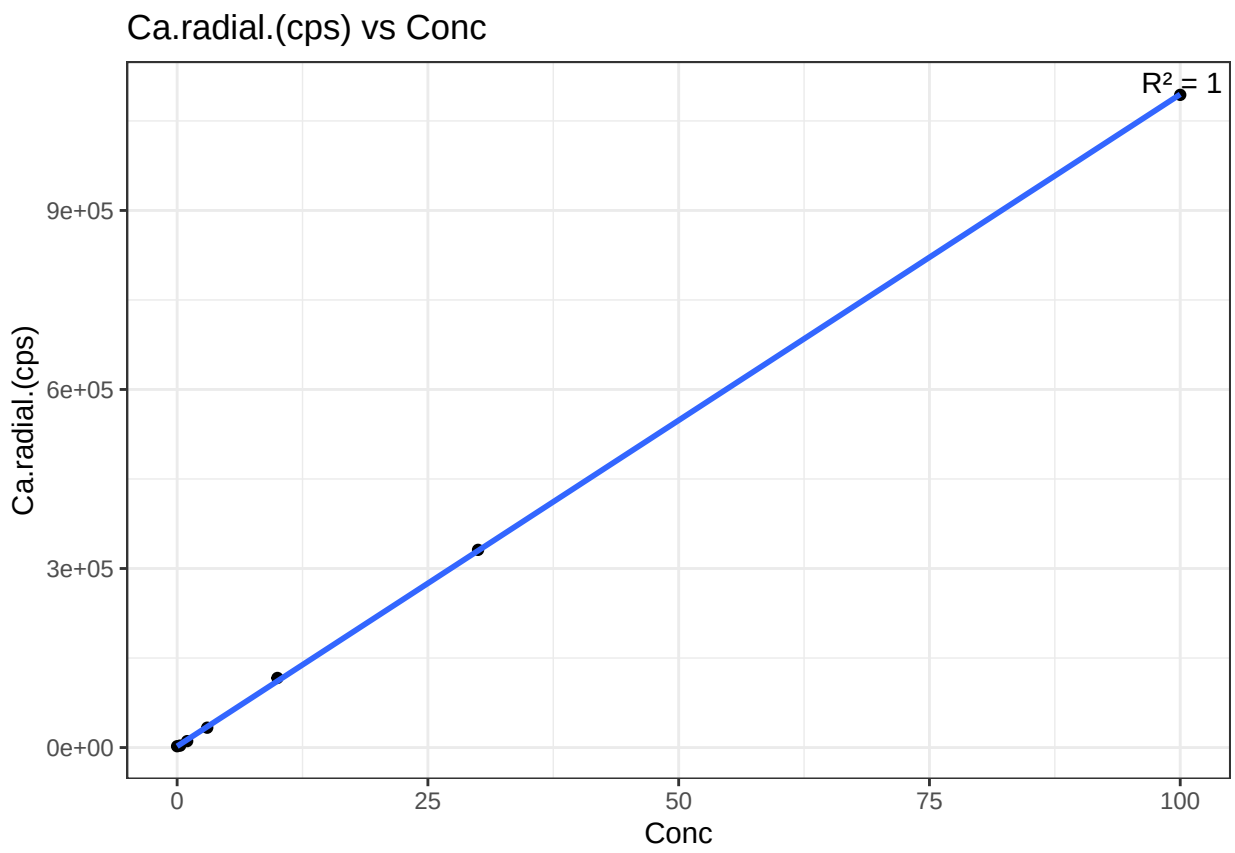

Mn.axial.(cps) vs Conc
R² = 1
Mn.axial.(cps)
4e+06
3e+06
2e+06
1e+06
0e+00
0
25
50
75
100
Conc

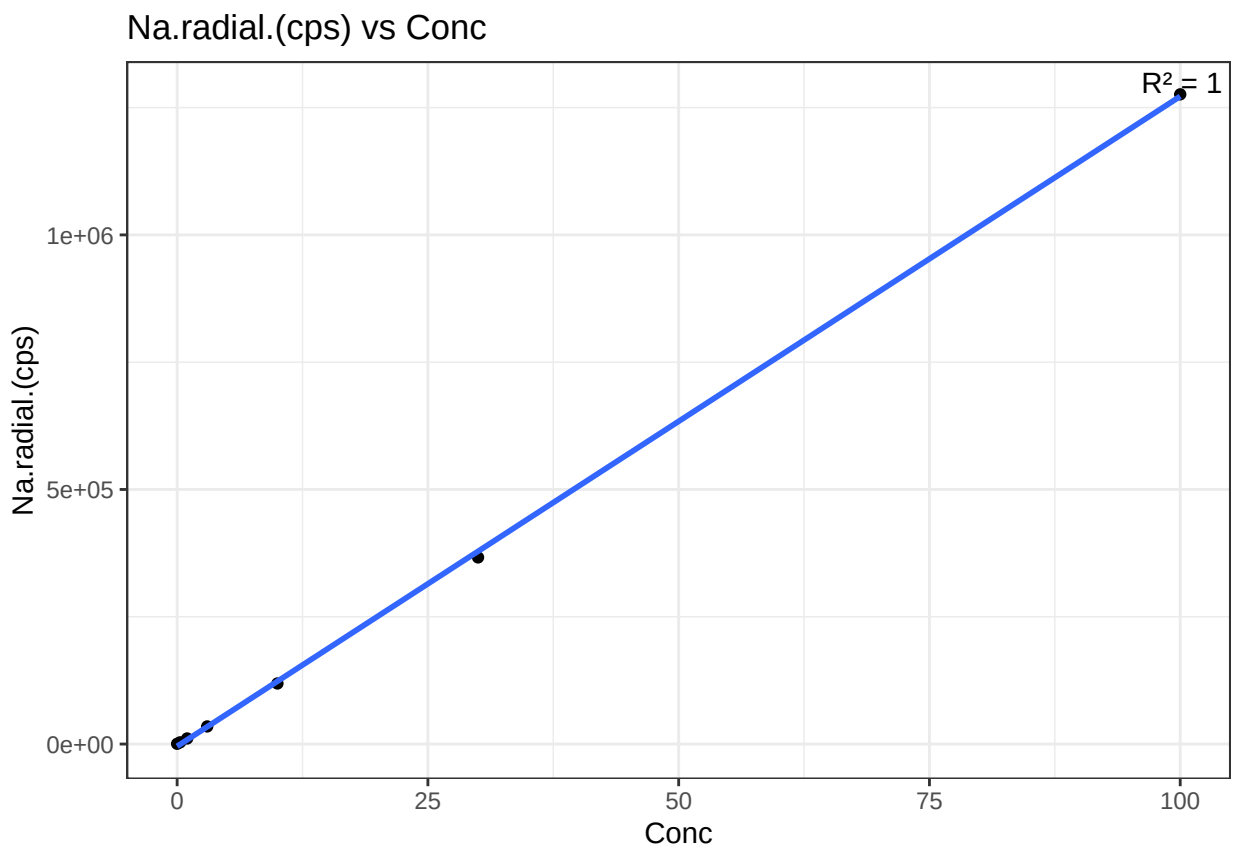

Al.axial.(cps) vs Conc
R² = 1
4e+06
3e+06
2e+06
1e+06
0e+00
Al.axial.(cps)
0
25
50
75
100
Conc


Sr.radial.(cps) vs Conc
R² = 0.912
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
Sr.radial.(cps)
0
25
50
75
100
Conc


Mg.radial.(cps) vs Conc
R² = 1
Mg.radial.(cps)
2000000
1500000
1000000
500000
0
0
25
50
75
100
Conc

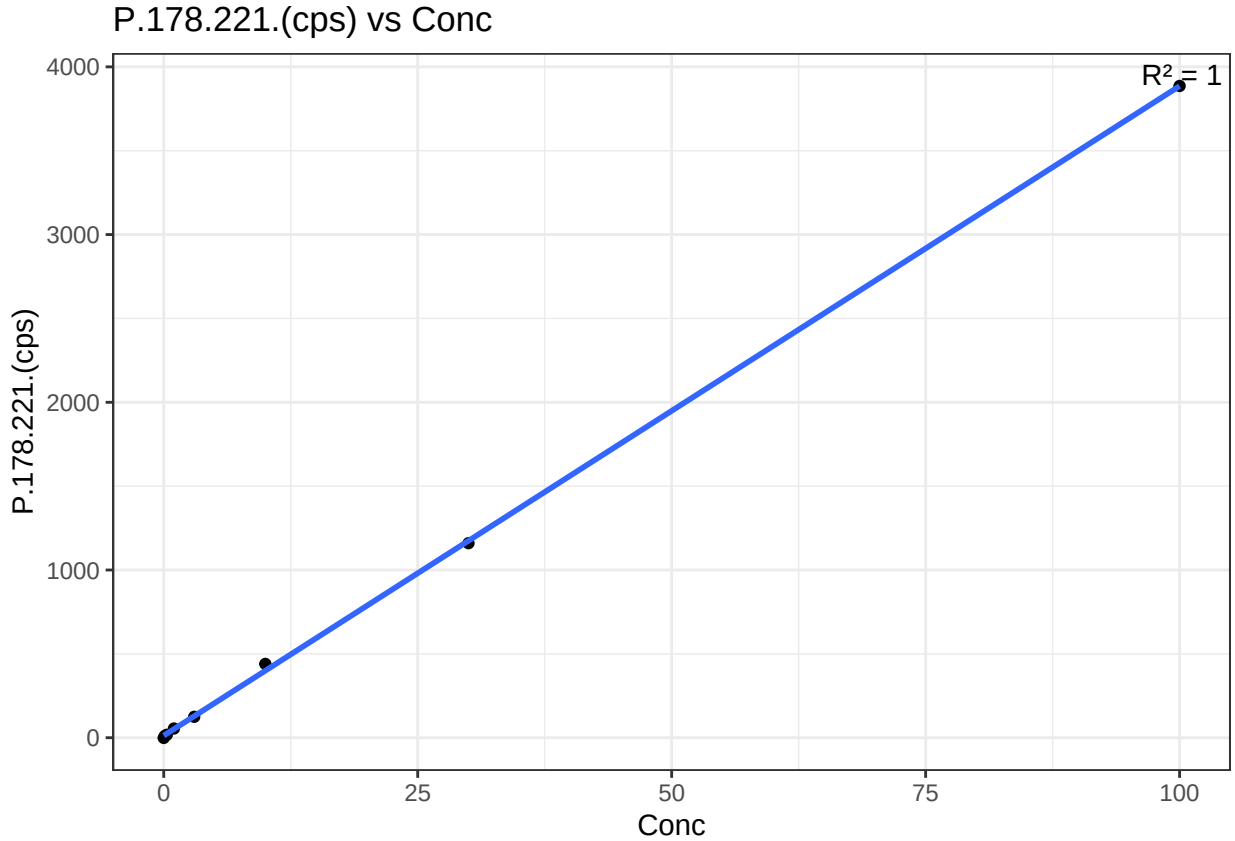

K.radial.(cps) vs Conc
R² = 0.858
500
250
0
−250
K.radial.(cps)
0
25
50
75
100
Conc

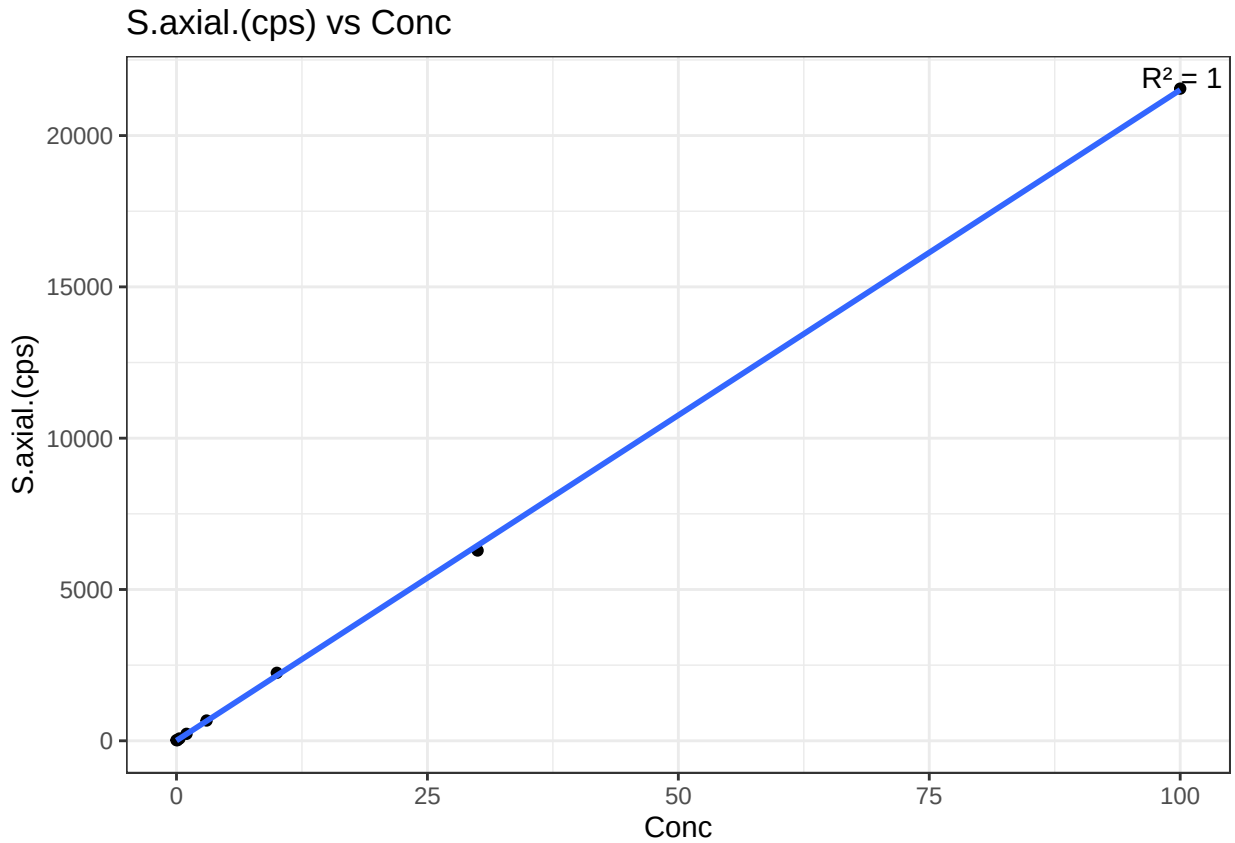

Ca.radial.(cps) vs Conc
R² = 1
9e+05
6e+05
3e+05
0e+00
Ca.radial.(cps)
0
25
50
75
100
Conc

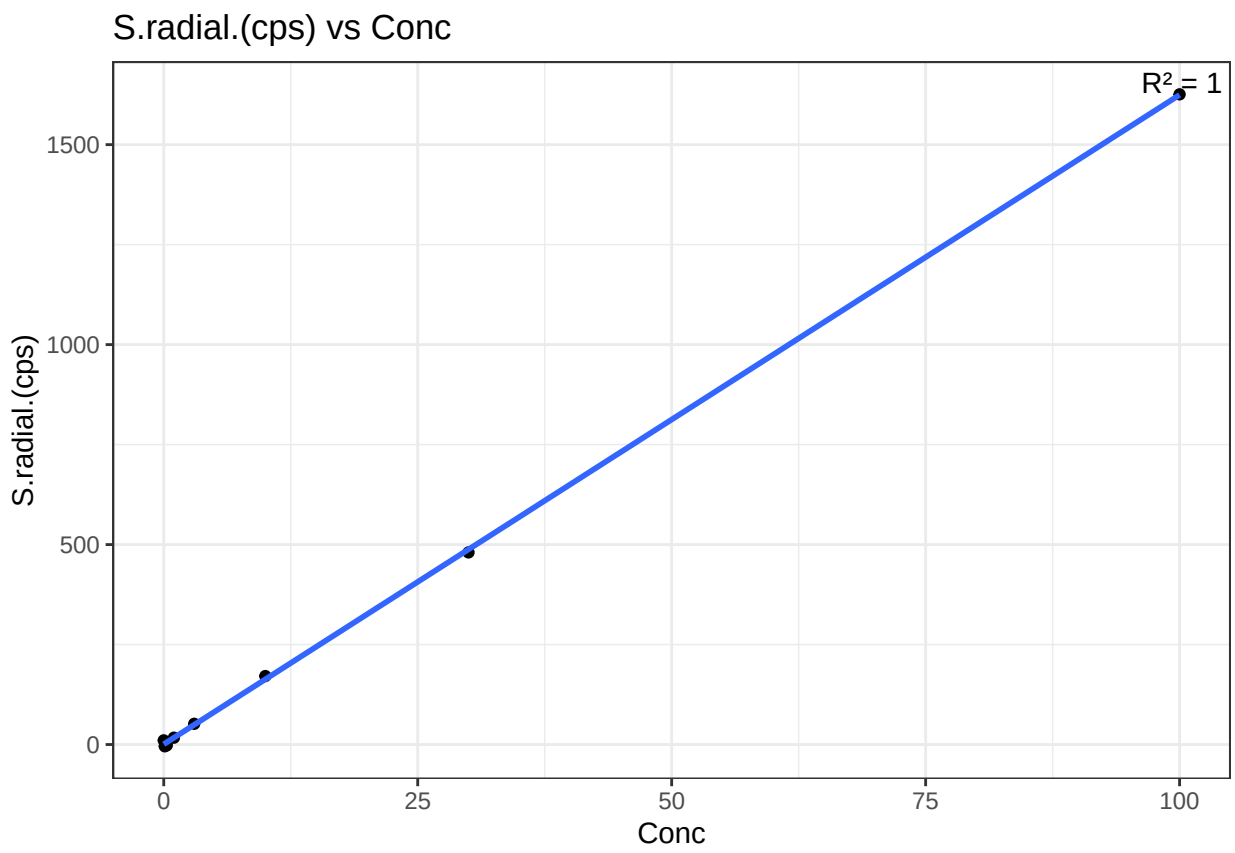


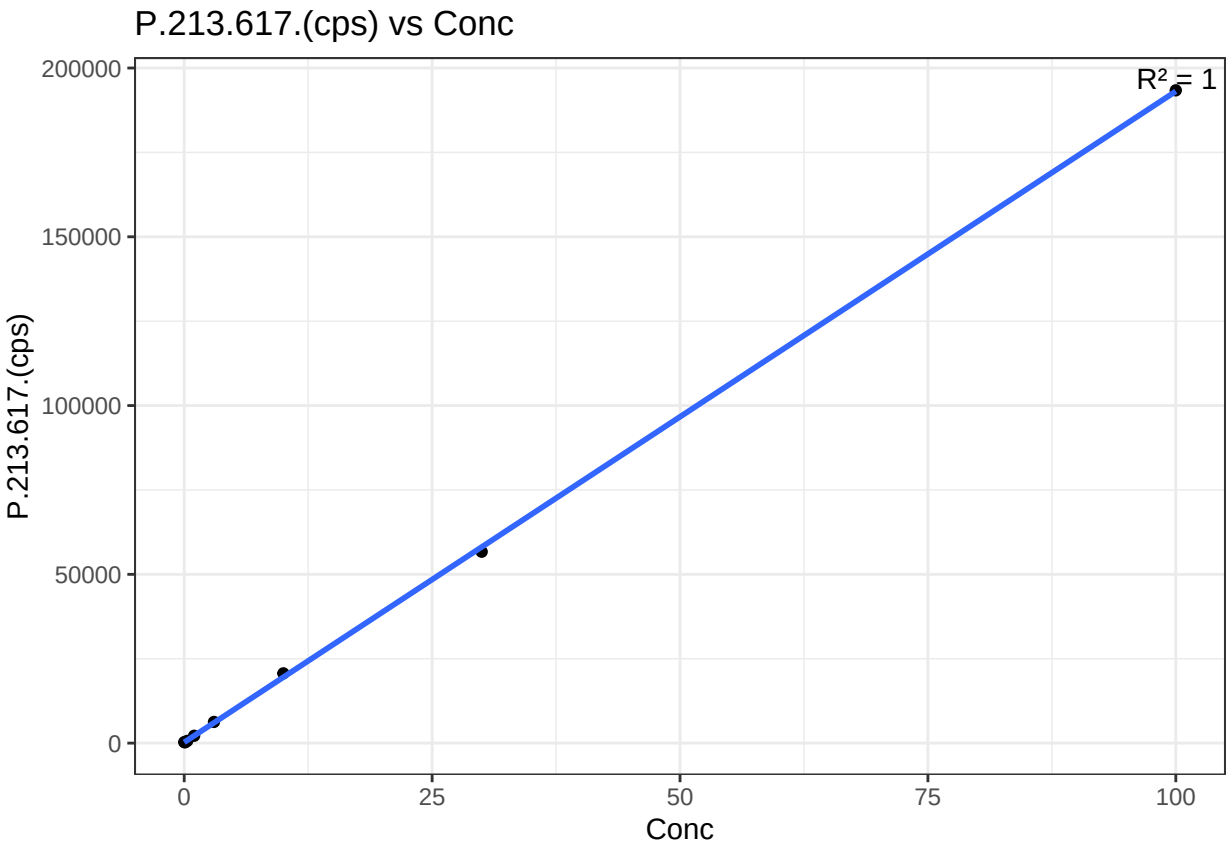

```
##           Variable        R2        Slope     Intercept
## 1  Zn.radial.(cps) 0.9996910 4.434816e+03  9.168948e+02
## 2  Ba.radial.(cps) 0.9997865 5.863004e+03  2.195985e+02
## 3  Fe.radial.(cps) 0.9992605 6.108641e+03 -2.687996e+03
## 4   Fe.axial.(cps) 0.9990945 4.095201e+04 -1.679934e+04
## 5    B.axial.(cps) 0.9999968 3.527538e+04 -2.918539e+02
## 6   B.radial.(cps) 0.9999775 8.144866e+03  2.006840e+01
## 7   Mn.axial.(cps) 0.9998091 4.471049e+05  9.496773e+03
## 8   Al.axial.(cps) 0.9997703 3.992801e+04 -1.130328e+04
## 9  Sr.radial.(cps) 0.9121837 9.649745e+09 -4.384256e+10
## 10 Mg.radial.(cps) 0.9998535 2.313379e+04 -4.892766e+03
## 11  K.radial.(cps) 0.8576786 2.238686e+01 -4.498415e+01
## 12 Ca.radial.(cps) 0.9999550 1.092481e+04  2.067301e+03
## 13 Na.radial.(cps) 0.9998274 1.276529e+04 -4.437924e+03
```

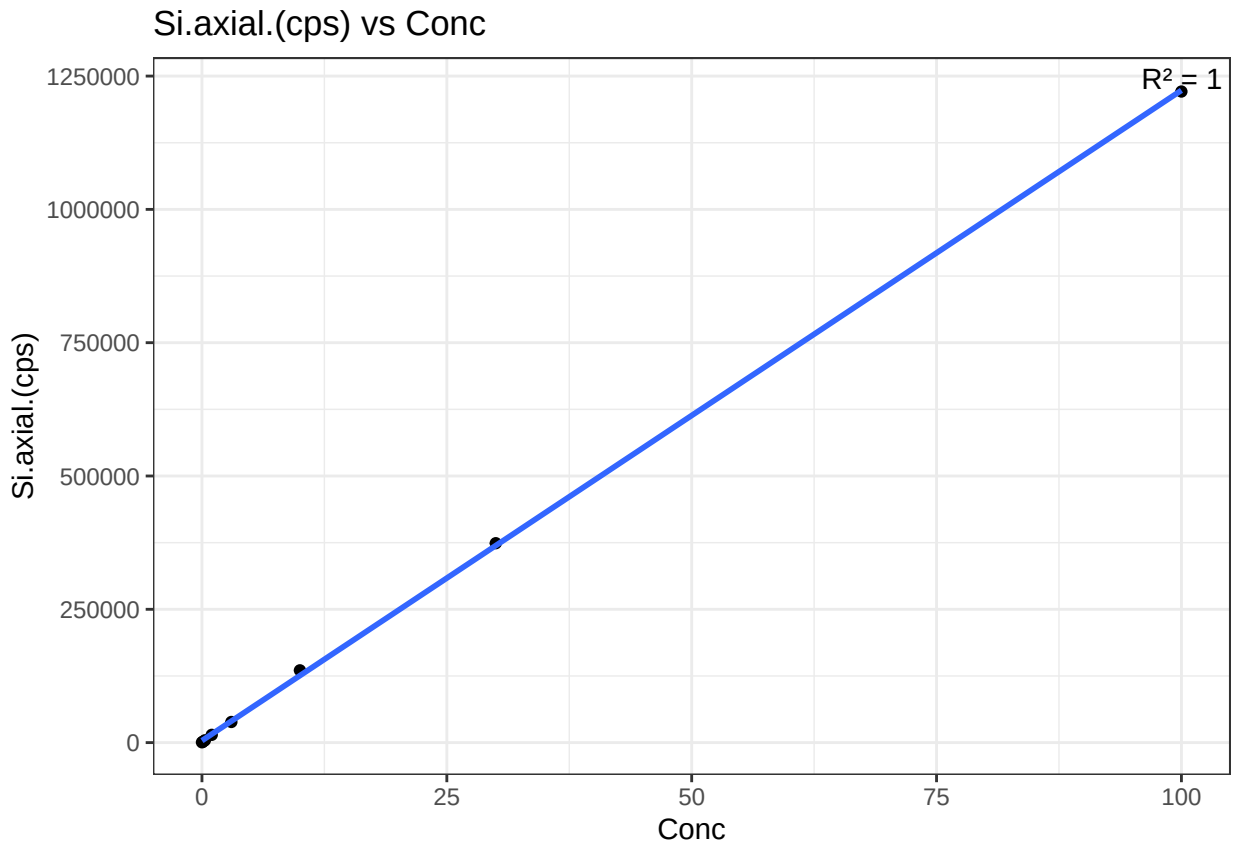
## 0.2 Import and QAQC S/P/Si Standards




S.radial.(cps) vs Conc
R² = 1
1500
1000
500
0
S.radial.(cps)
0
25
50
75
100
Conc

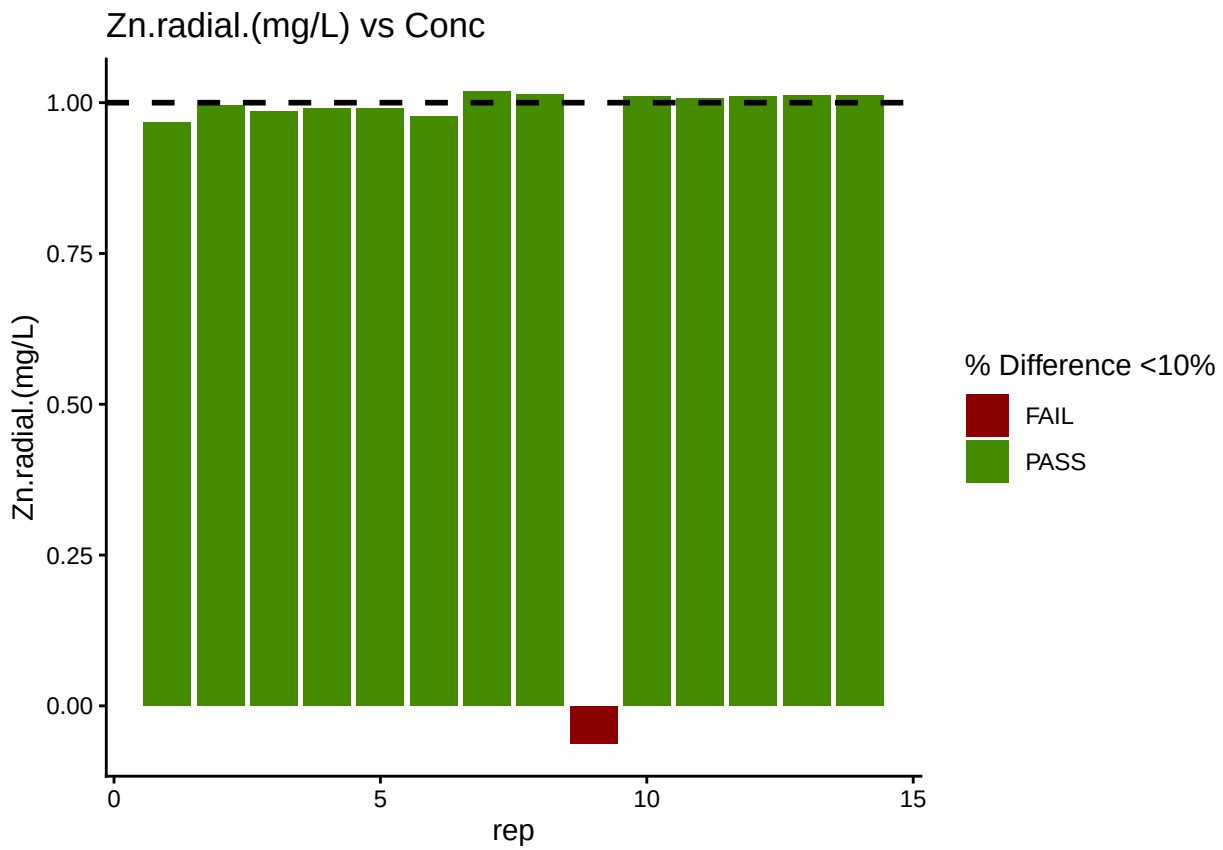

P.213.617.(cps) vs Conc
R² = 1
200000
150000
100000
50000
0
P.213.617.(cps)
0
25
50
75
100
Conc



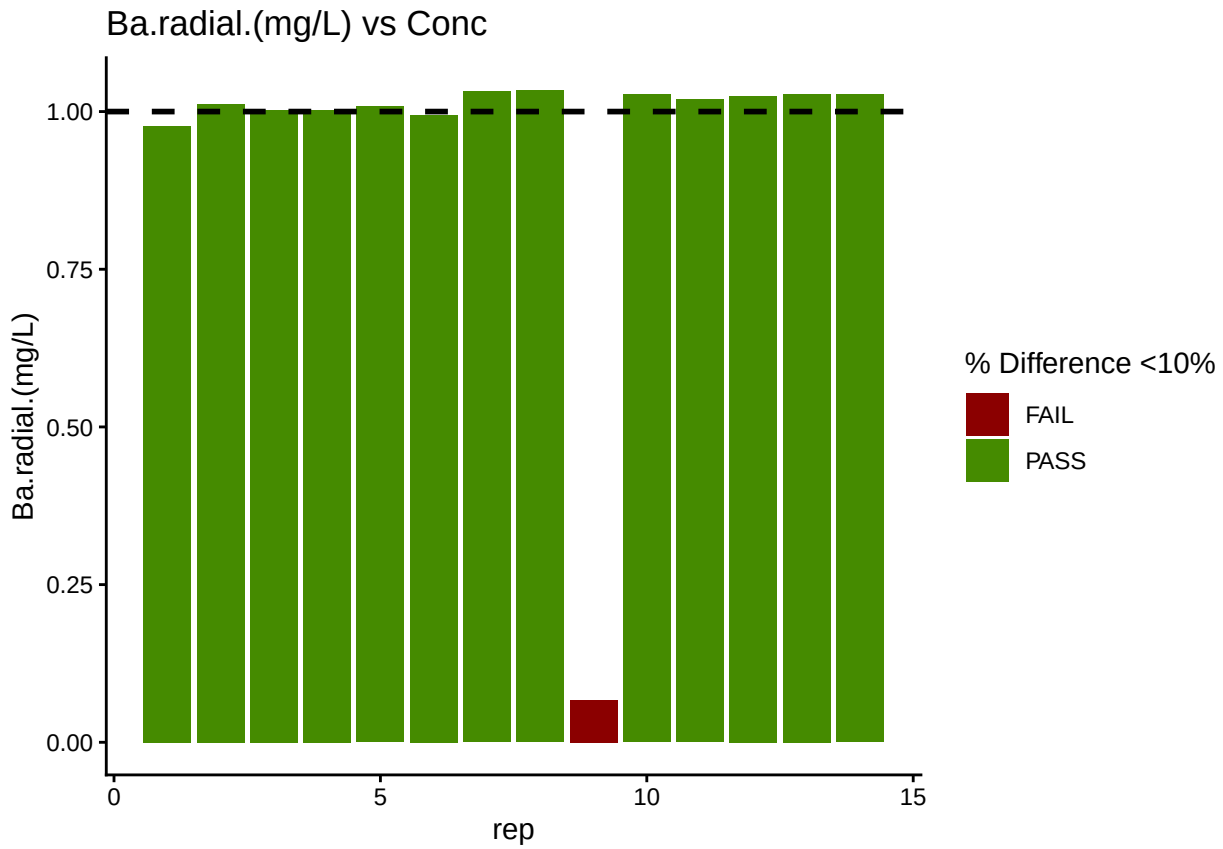

```
##          Variable        R2       Slope    Intercept
## 1 P.178.221.(cps) 0.9998313    38.73282   12.3597996
## 2   S.axial.(cps) 0.9998991   215.03204    6.6526079
## 3  S.radial.(cps) 0.9998528    16.23644    0.7511049
## 4 P.213.617.(cps) 0.9998890  1928.54387  237.5838996
## 5  Si.axial.(cps) 0.9998743 12196.23912 3651.0837126
```

##Import Sample Data

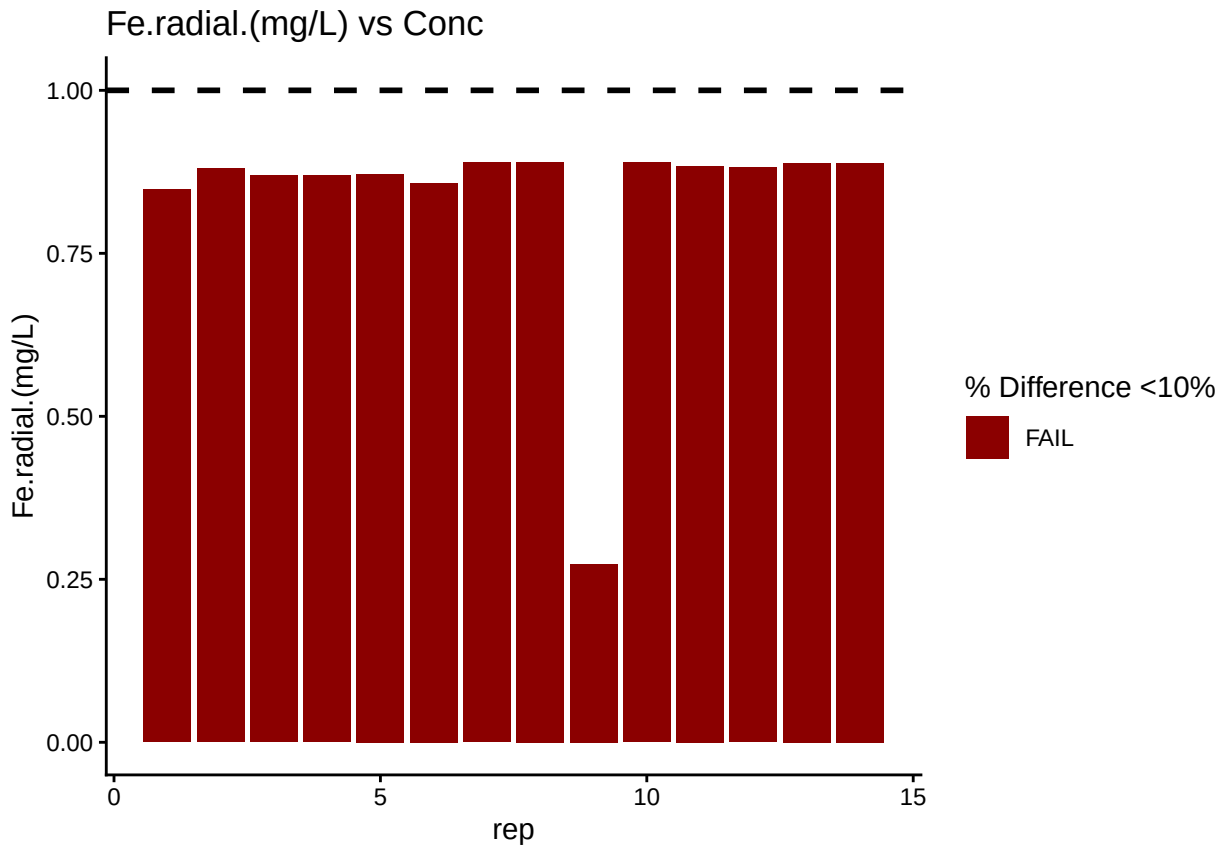
## 0.3 Assess Metal Check Standards



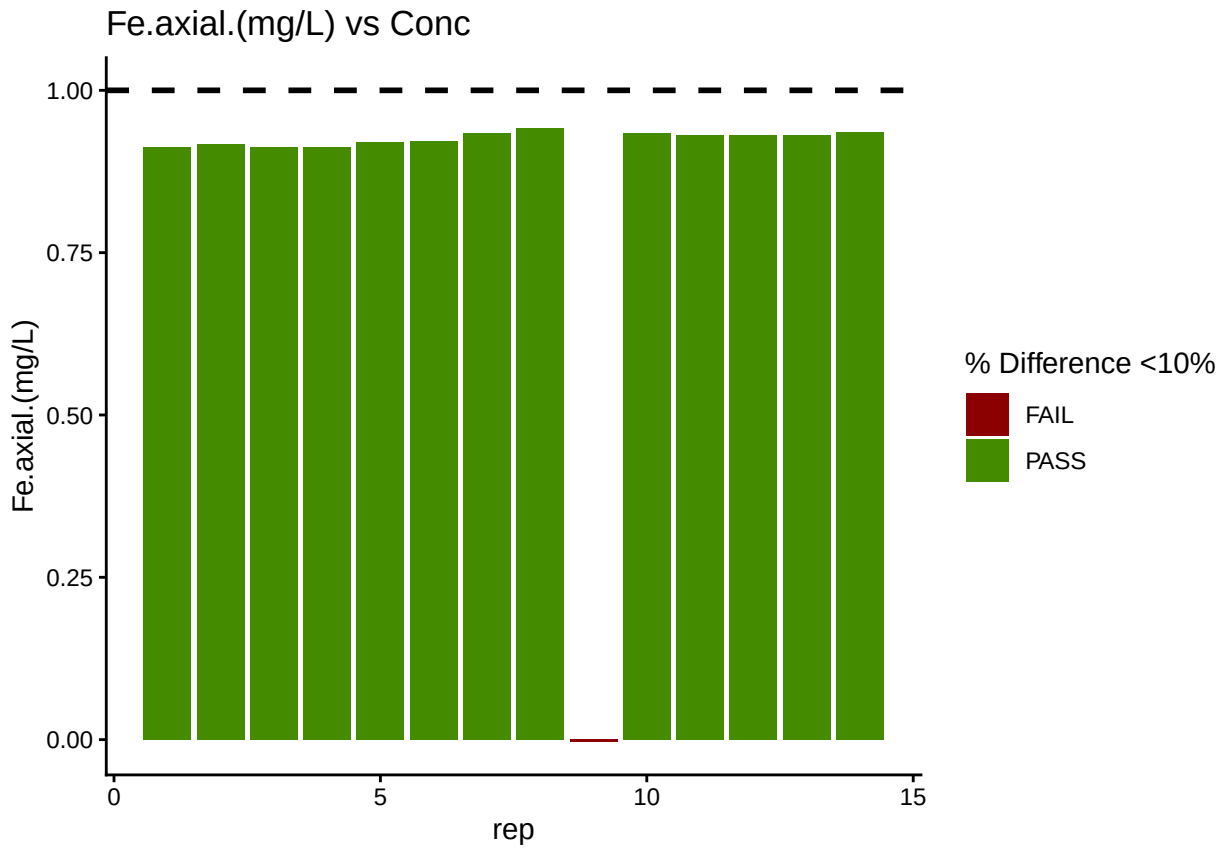

```
## [1] ">80% of Zn.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Zn.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



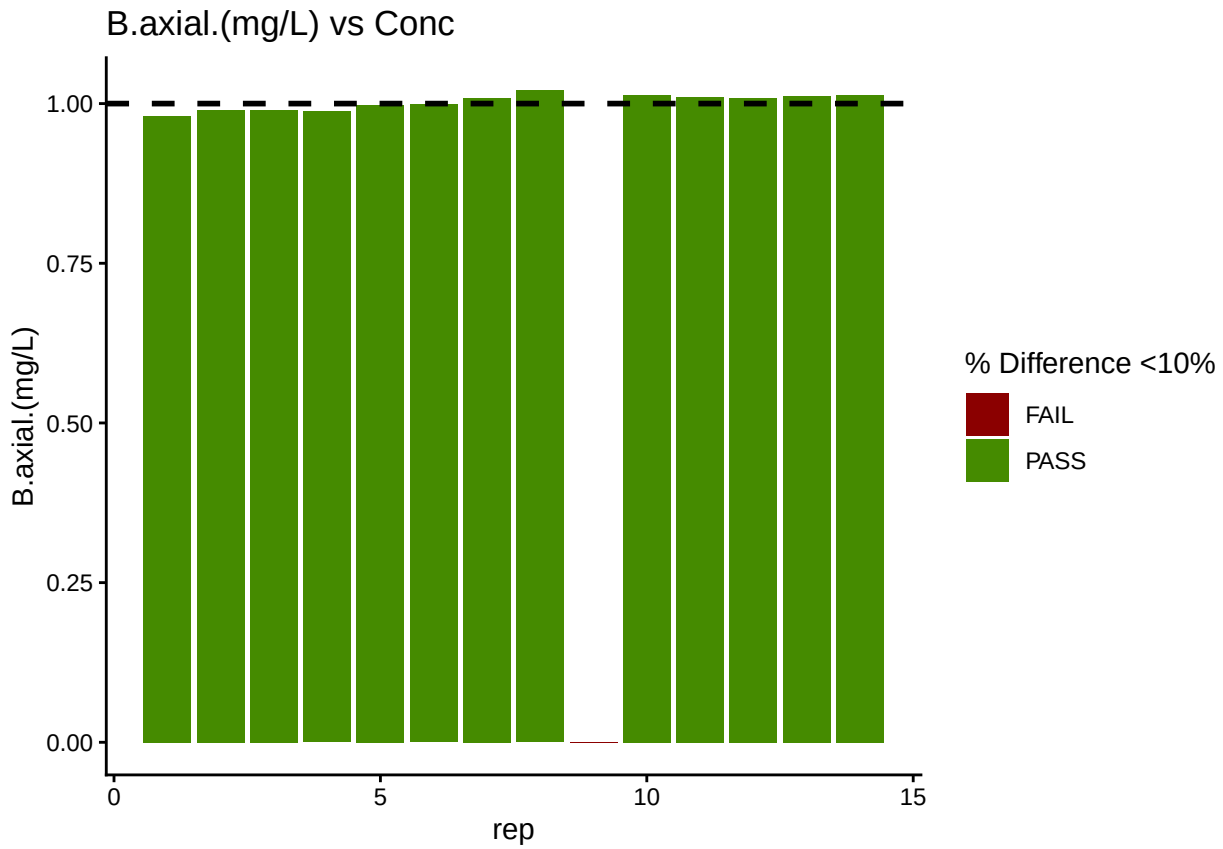

```
## [1] ">80% of Ba.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Ba.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



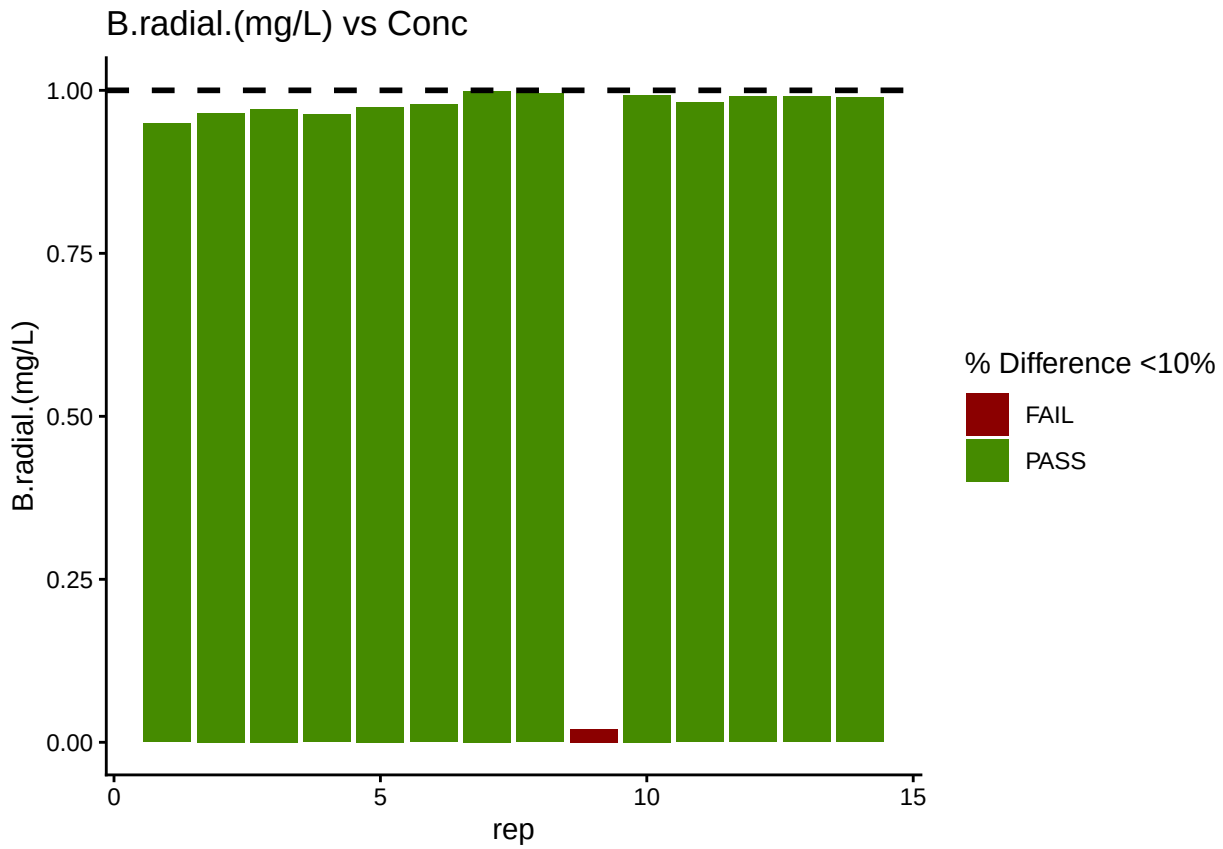

```
## [1] "<80% of Fe.radial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "Fe.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



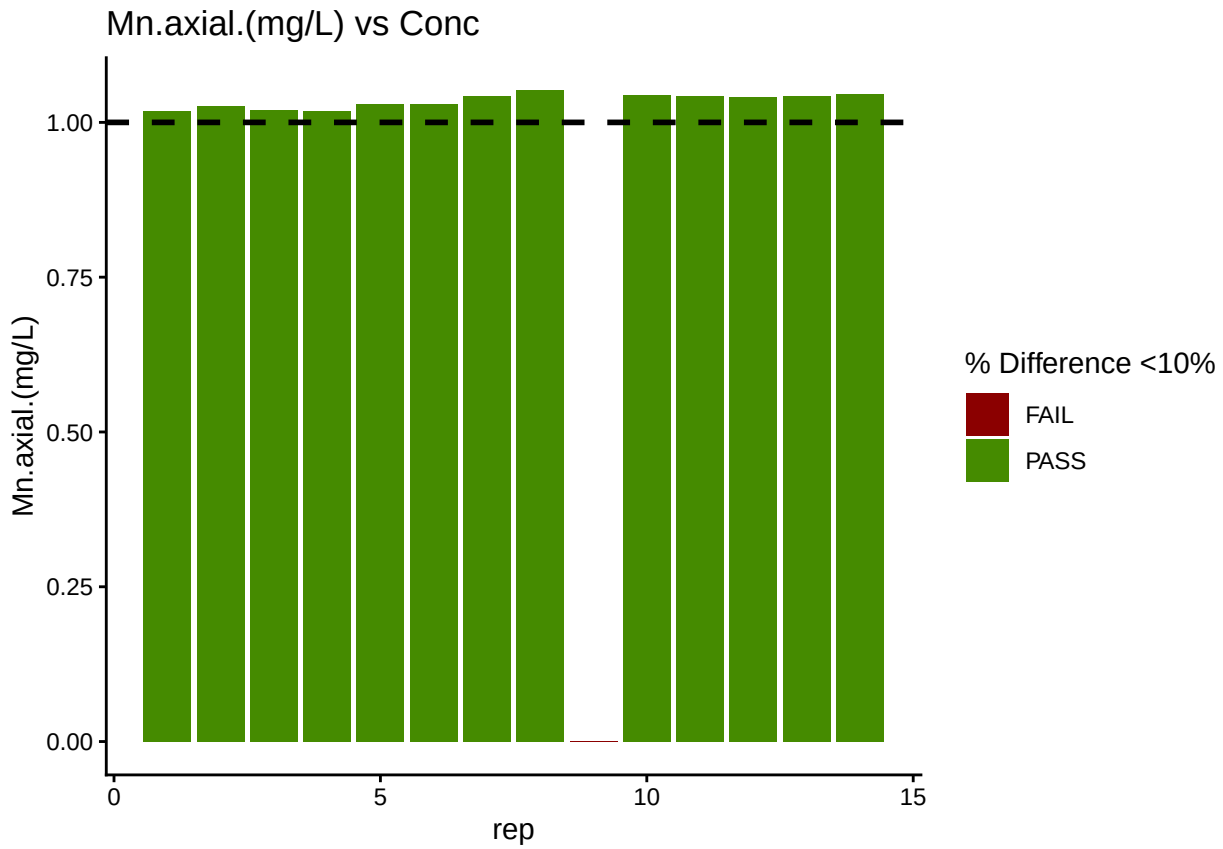

```
## [1] ">80% of Fe.axial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Fe.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



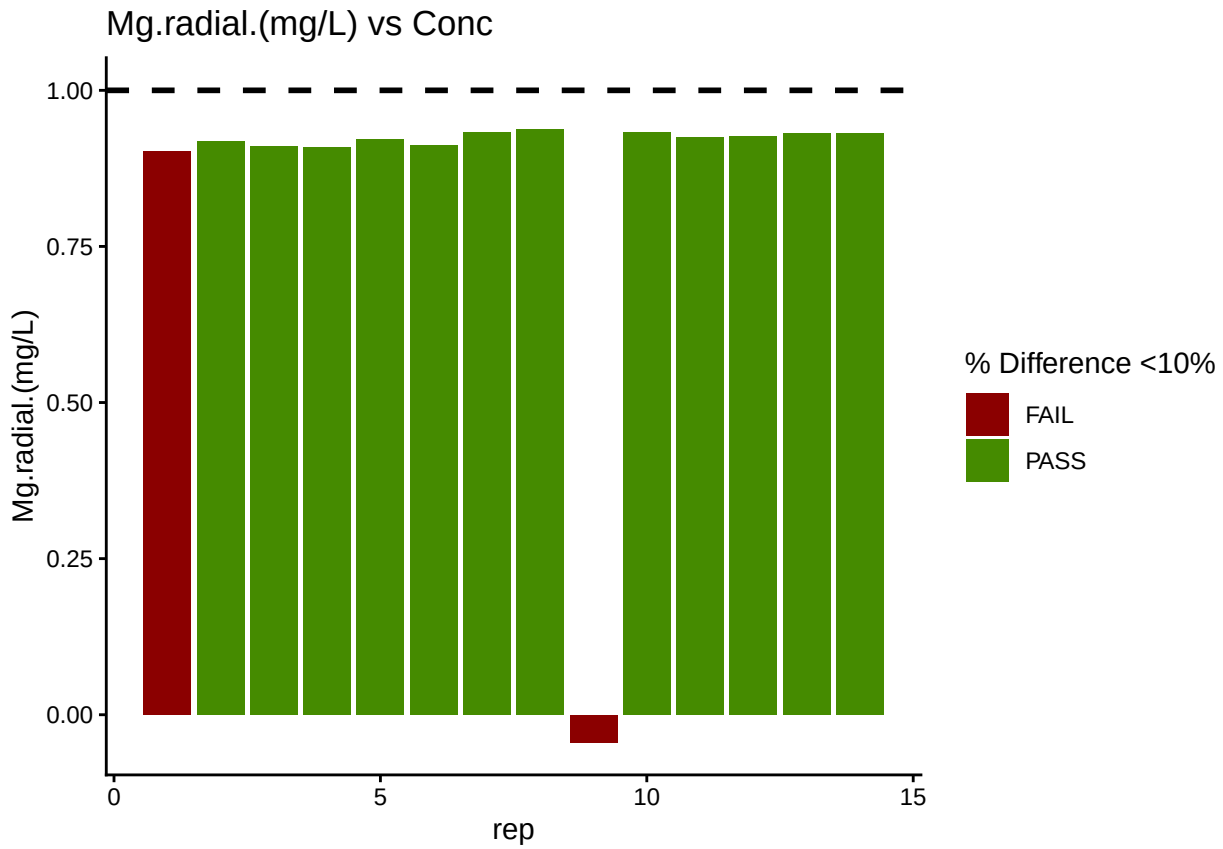

```
## [1] ">80% of B.axial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "B.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



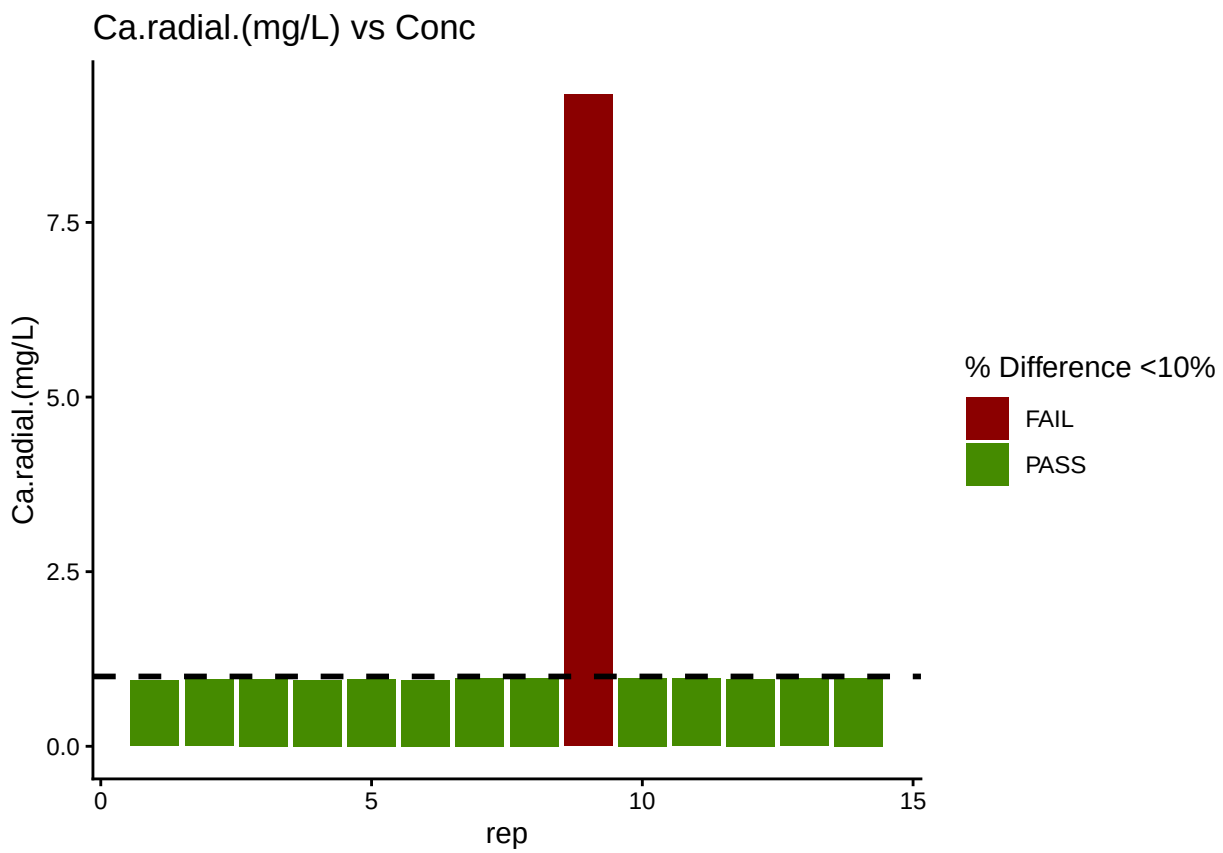

```
## [1] ">80% of B.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "B.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



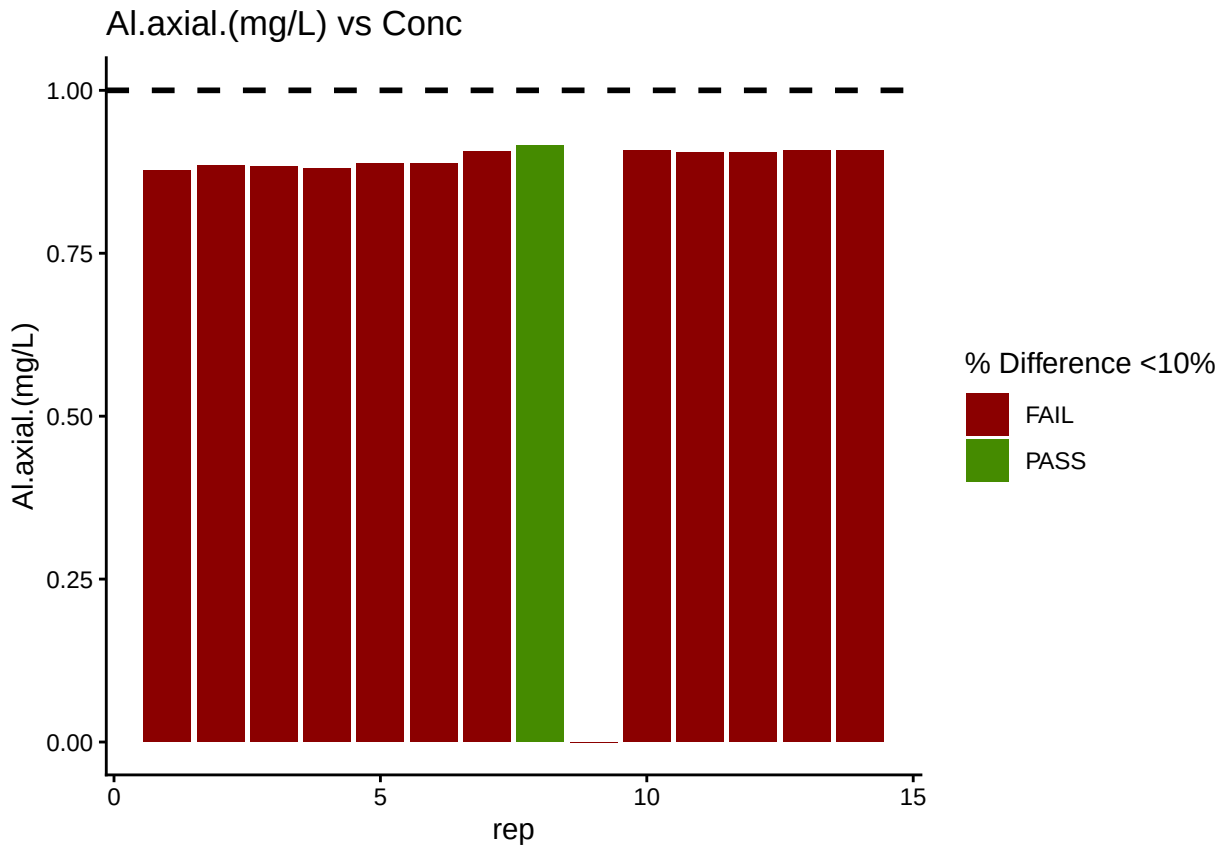

```
## [1] ">80% of Mn.axial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Mn.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



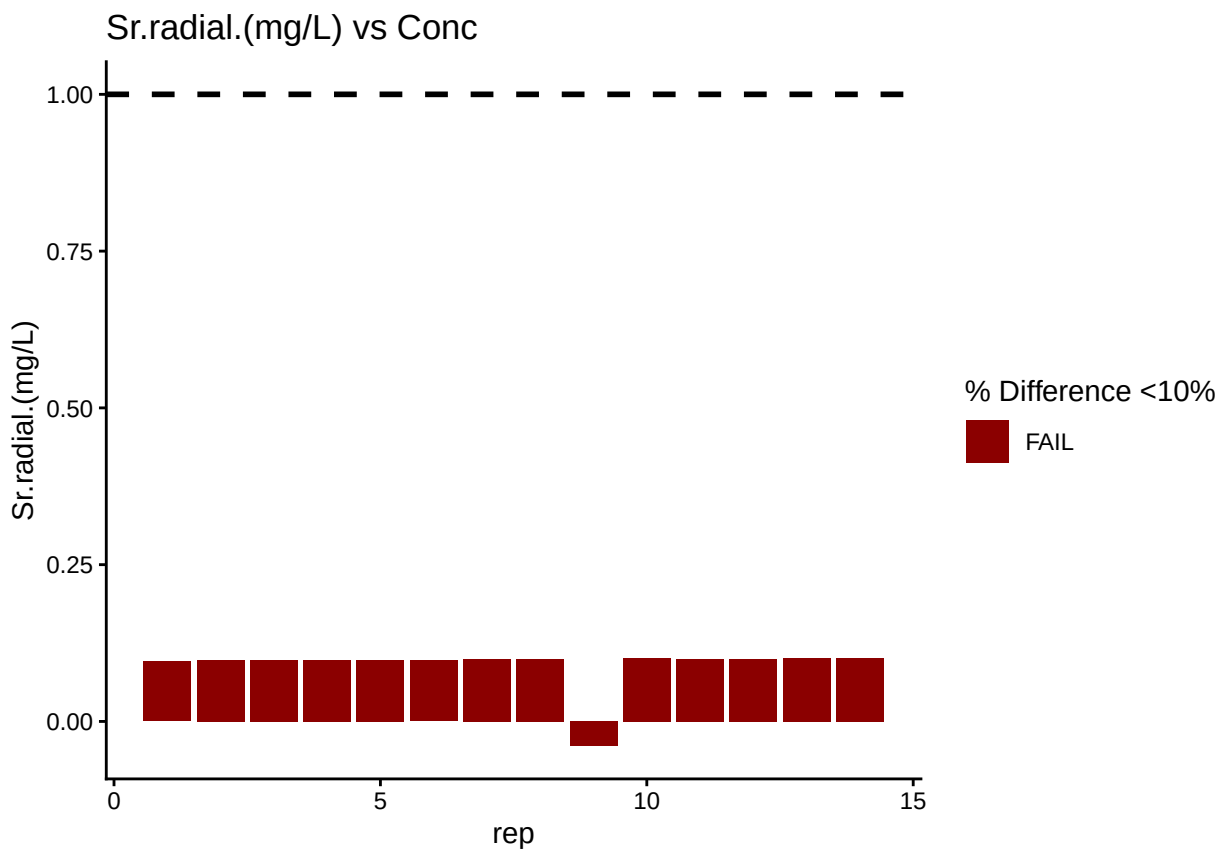

```
## [1] ">80% of Mg.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Mg.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



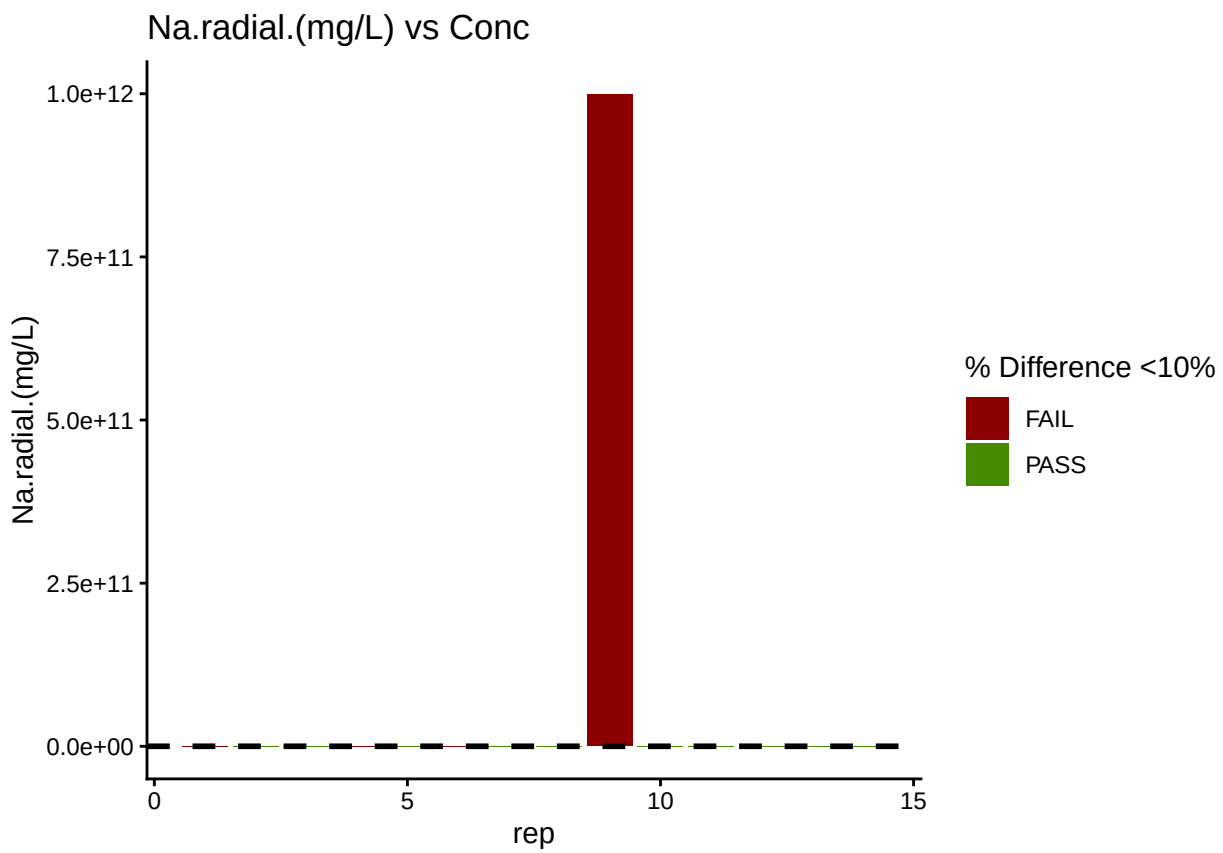

```
## [1] ">80% of Ca.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Ca.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



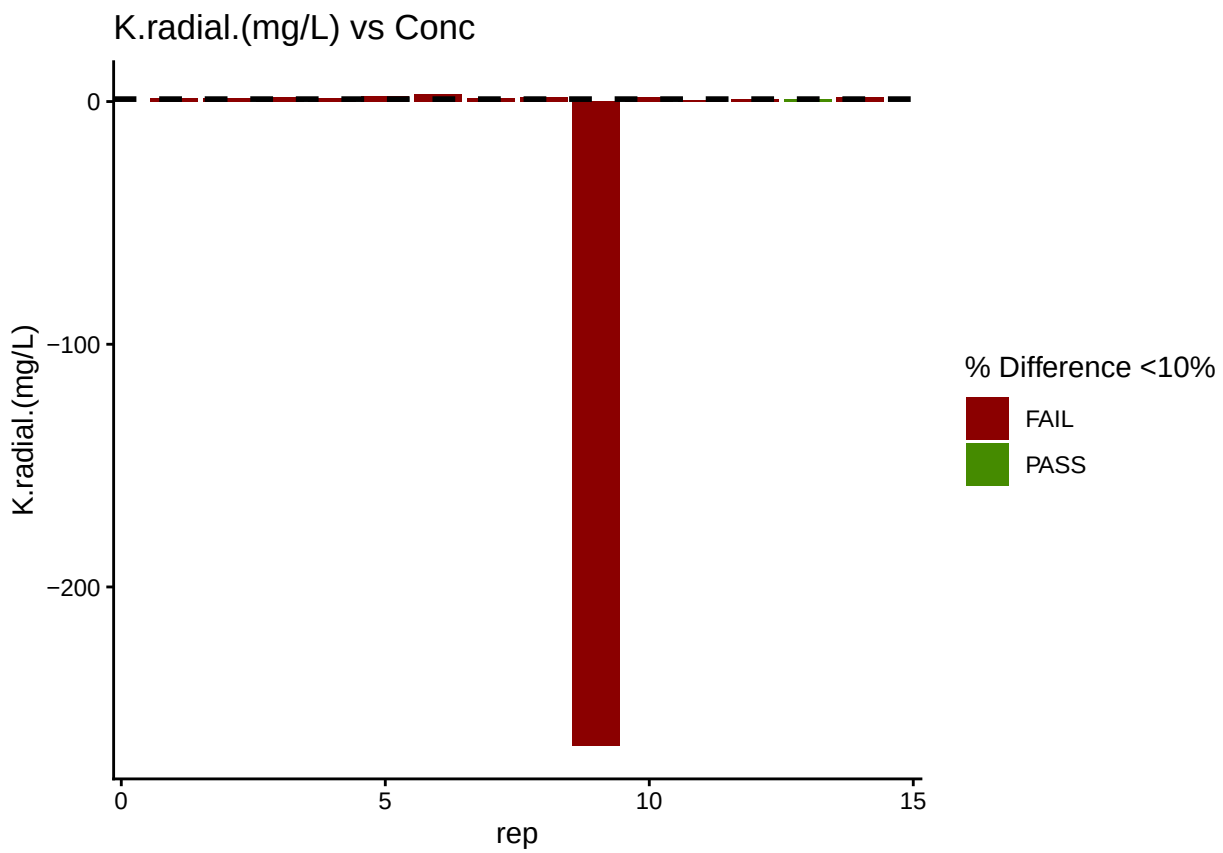

```
## [1] "<80% of Al.axial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "Al.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```




```
## [1] "<80% of Sr.radial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "Sr.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```

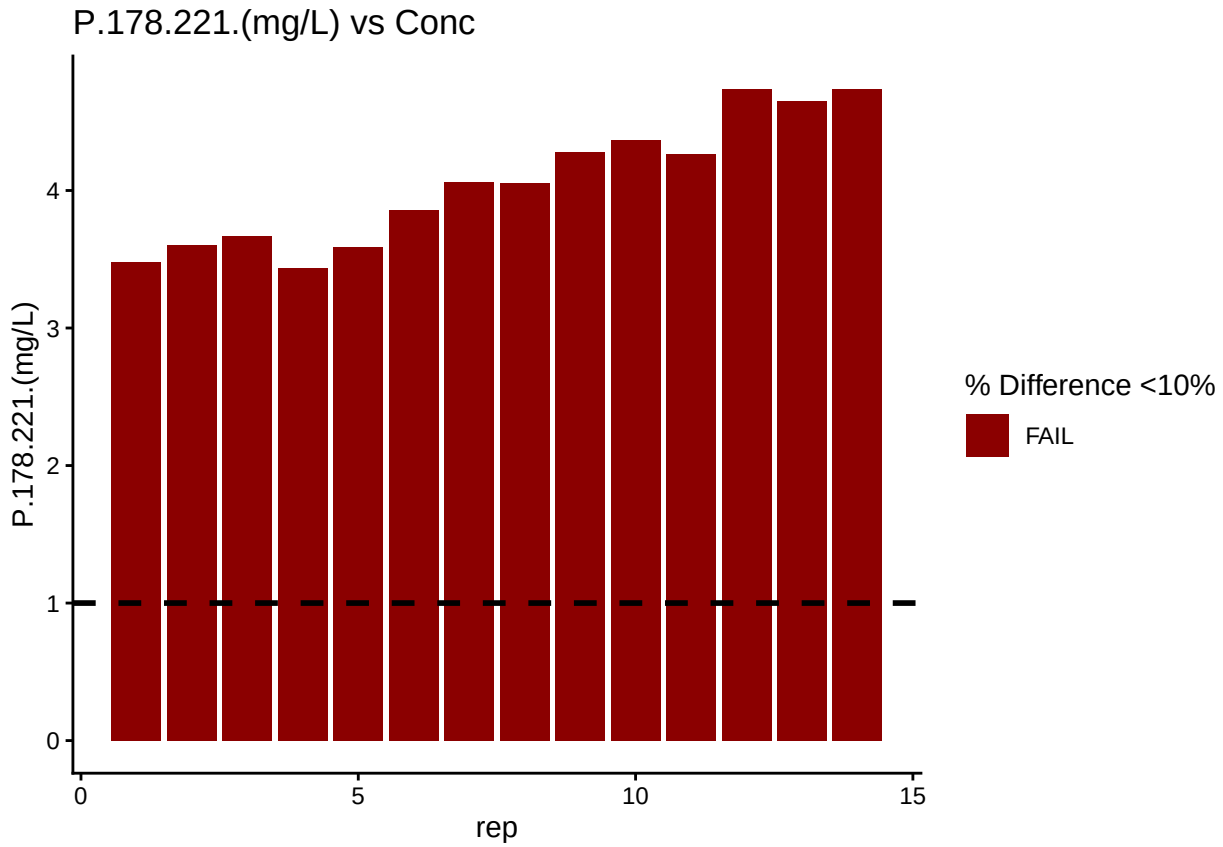



```
## [1] "<80% of Na.radial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "Na.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```

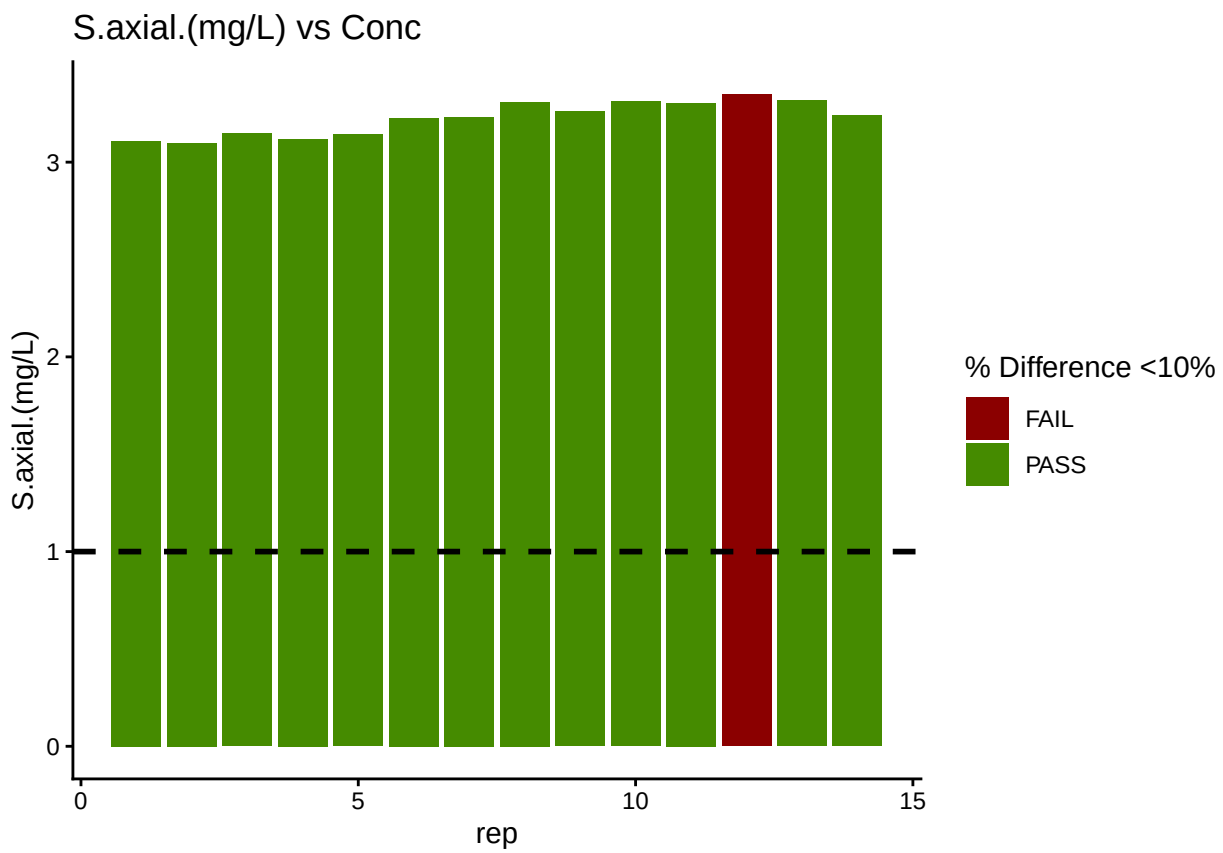



```
## [1] "<80% of K.radial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "K.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```

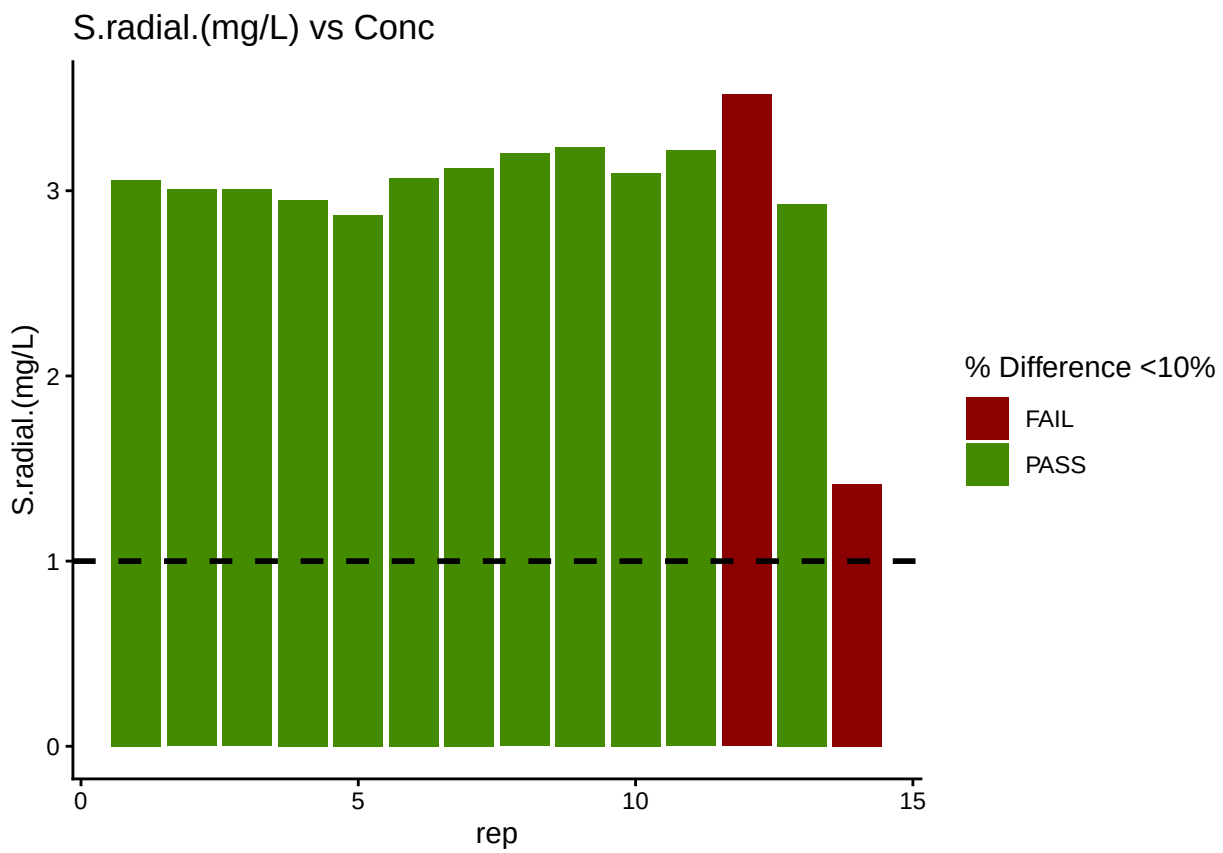
## 0.4 Assess S/P/Si Check Standards



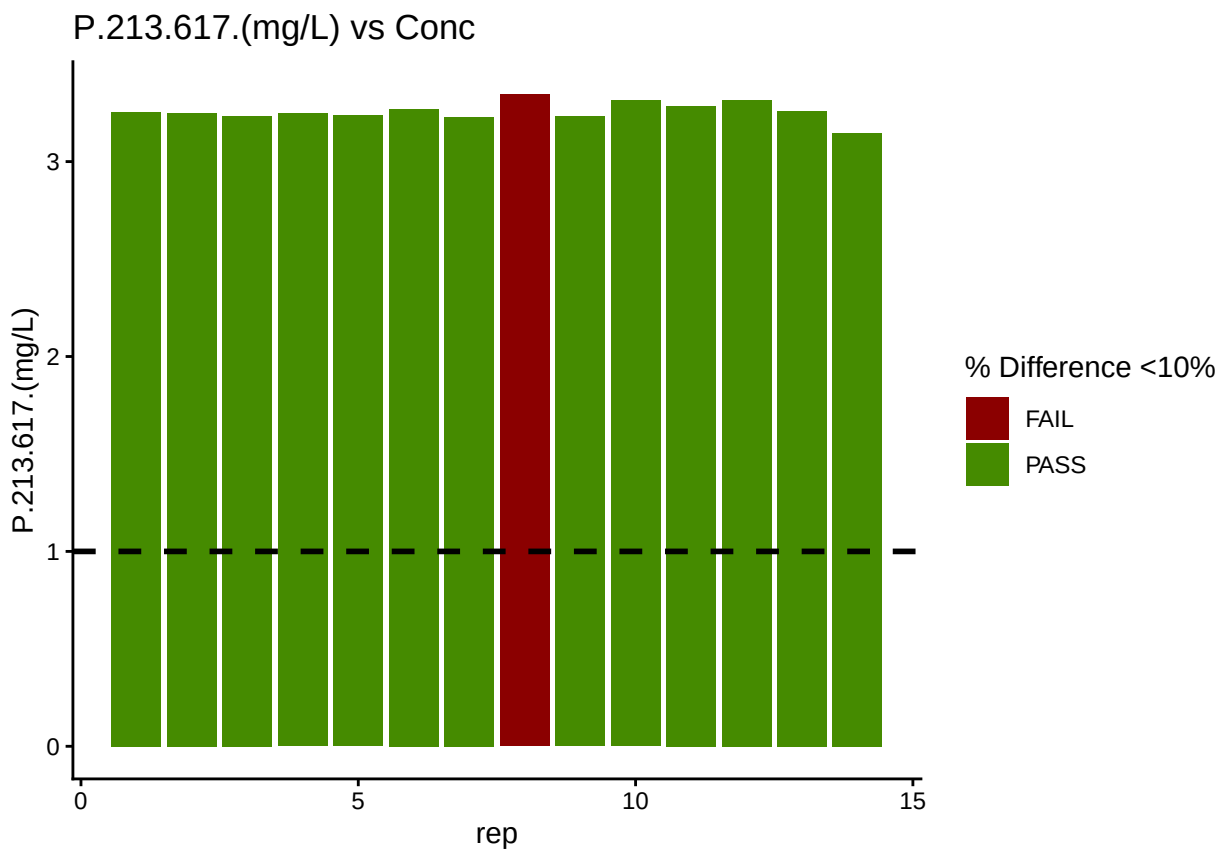

```
## [1] "<80% of P.178.221.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "P.178.221.(mg/L) RSV WITHIN RANGE - PROCEED"
```



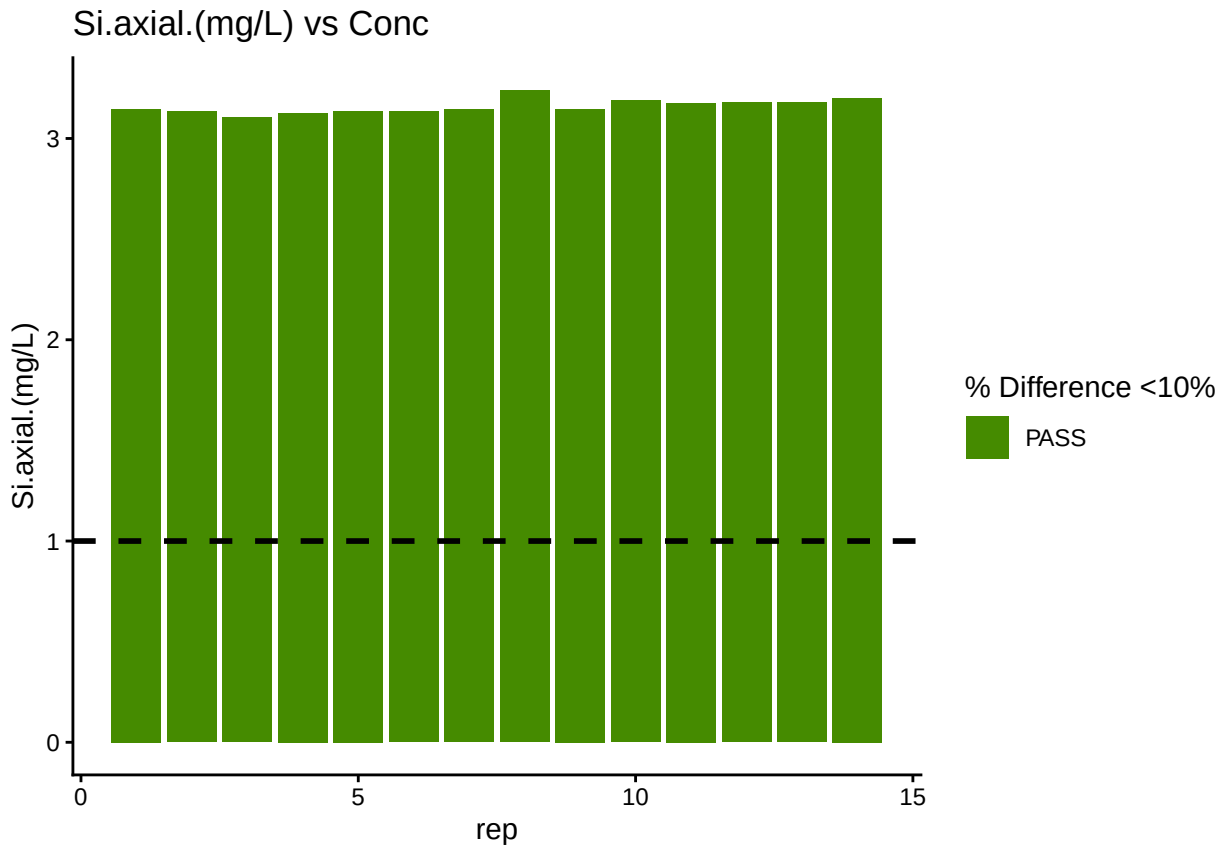

```
## [1] ">80% of S.axial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "S.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```



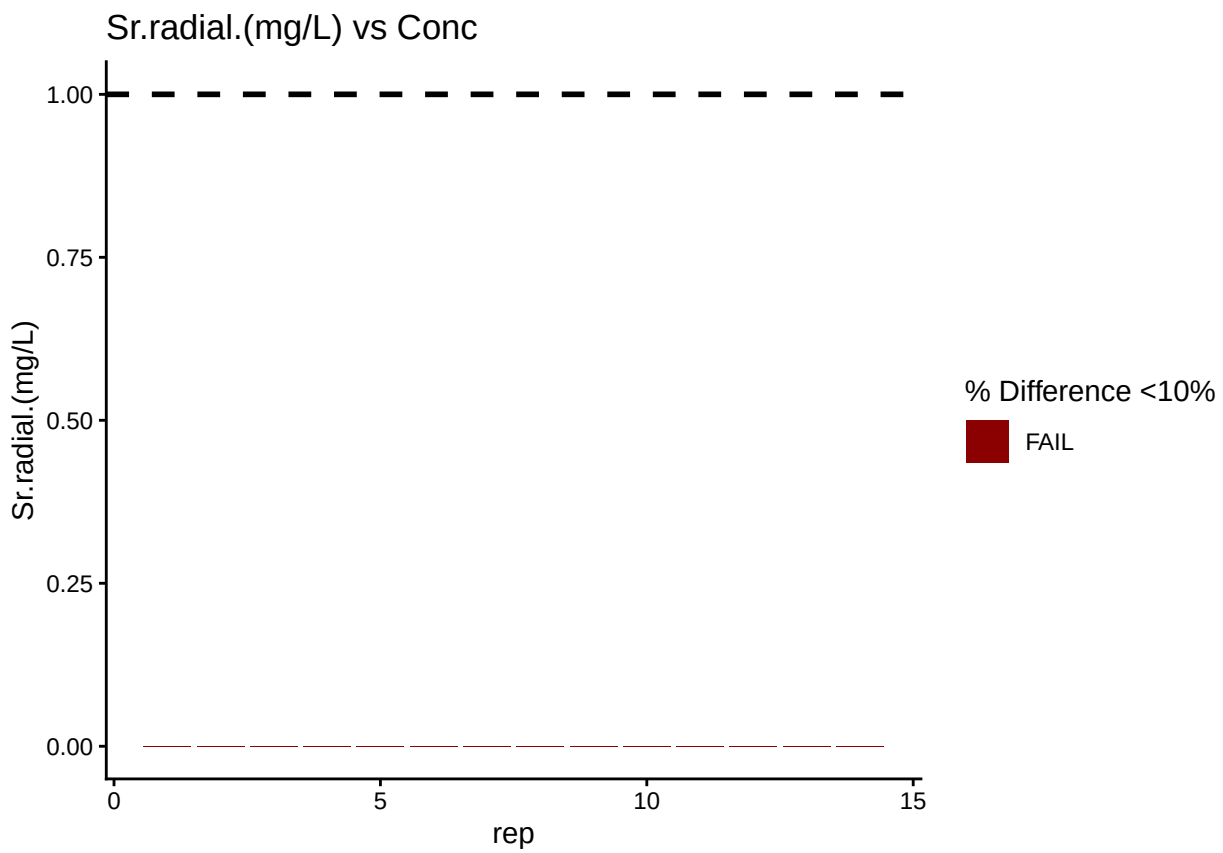

```
## [1] ">80% of S.radial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "S.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```




```
## [1] ">80% of P.213.617.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "P.213.617.(mg/L) RSV WITHIN RANGE - PROCEED"
```




```
## [1] ">80% of Si.axial.(mg/L) Check Standards are within range of expected concentration - PROCEED"
## [1] "Si.axial.(mg/L) RSV WITHIN RANGE - PROCEED"
```




```
## [1] "<80% of Sr.radial.(mg/L) Check Standards are within range of expected concentration - REASSESS"
## [1] "Sr.radial.(mg/L) RSV WITHIN RANGE - PROCEED"
```